

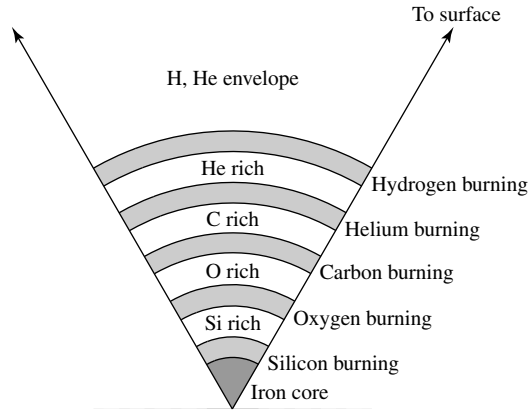


FIGURE 15.10 The onion-like interior of a massive star that has evolved through core silicon burning. Inert regions of processed material are sandwiched between the nuclear burning shells. The inert regions exist because the temperature and density are not sufficient to cause nuclear reactions to occur with that composition. (This drawing is not to scale.)

and cannot contribute to the luminosity of the star. Grouping all of the products together, silicon burning is said to produce an *iron core*. A sketch of the onion-like interior structure of a massive star following silicon burning is given in Fig. 15.10.

Because carbon, oxygen, and silicon burning produce nuclei with masses progressively nearer the *iron peak* of the binding energy curve, less and less energy is generated per unit mass of fuel. As a result, the timescale for each succeeding reaction sequence becomes shorter (recall Example 10.3.1). For example, for a $20 M_{\odot}$ star, the main-sequence lifetime (core hydrogen burning) is roughly 10^7 years, core helium burning requires 10^6 years, carbon burning lasts 300 years, oxygen burning takes roughly 200 days, and silicon burning is completed in only two days!

At the very high temperatures now present in the core, the photons possess enough energy to destroy heavy nuclei (note the reverse arrows in the silicon-burning sequence), a process known as **photodisintegration**. Particularly important are the photodisintegration of $^{56}_{26}\text{Fe}$ and ^4_2He :



When the mass of the contracting iron core has become large enough and the temperature sufficiently high, photodisintegration can, in a very short period of time, undo what the star has been trying to do its entire life, namely produce elements more massive than hydrogen and helium. Of course, this process of stripping iron down to individual protons and neutrons is highly endothermic, as suggested in Example 15.3.1; thermal energy is removed from the gas that would otherwise have resulted in the pressure necessary to support the core of the star. The core masses for which this process occurs vary from $1.3 M_{\odot}$ for a $10 M_{\odot}$ ZAMS star to $2.5 M_{\odot}$ for a $50 M_{\odot}$ star.

Under the extreme conditions that now exist (e.g., $T_c \sim 8 \times 10^9$ K and $\rho_c \sim 10^{13}$ kg m⁻³ for a 15 $M_\odot$ star), the free electrons that had assisted in supporting the star through degeneracy pressure are captured by heavy nuclei and by the protons that were produced through photodisintegration; for instance,



The amount of energy that escapes the star in the form of neutrinos becomes enormous; during silicon burning the photon luminosity of a 20 $M_\odot$ stellar model is 4.4×10^{31} W while the neutrino luminosity is 3.1×10^{38} W.

Through the photodisintegration of iron, combined with electron capture by protons and heavy nuclei, most of the core's support in the form of electron degeneracy pressure is suddenly gone and the core begins to collapse extremely rapidly. In the inner portion of the core, the collapse is homologous, and the velocity of the collapse is proportional to the distance away from the center of the star (recall Eq. 12.26 and the discussion of a homologous free-fall collapse during the formation of a protostar). At the radius where the velocity exceeds the local sound speed, the collapse can no longer remain homologous and the inner core decouples from the now supersonic outer core, which is left behind and nearly in free-fall. During the collapse, speeds can reach almost 70,000 km s⁻¹ in the outer core, and within about one second a volume the size of Earth has been compressed down to a radius of 50 km!

Example 15.3.2. If a mass with the radius of Earth ($R_\oplus$) collapses to a radius of only 50 km, a tremendous amount of gravitational potential energy would be released. Can this energy release be responsible for the energy of a core-collapse supernova?

Assume for simplicity that we can use Newtonian physics to estimate the amount of energy released during the collapse. From the virial theorem (see Eq. 10.23), the energy released in the formation of a spherically symmetric star of constant density is

$$E \sim -\frac{3}{10} \frac{GM^2}{R}.$$

Equating the energy of a Type II supernova, $E_{\text{II}} = 10^{46}$ J, to the gravitational energy released during the collapse, and given that $R_f = 50$ km $\ll R_\oplus$, the amount of mass required to produce the supernova would be

$$M \simeq \sqrt{\frac{10}{3} \frac{E_{\text{II}} R_f}{G}} \simeq 5 \times 10^{30} \text{ kg} \simeq 2.5 M_\odot.$$

This value is characteristic of the core masses mentioned earlier.

Since mechanical information will propagate through the star only at the speed of sound and because the core collapse proceeds so quickly, there is not enough time for the outer layers to learn about what has happened inside. The outer layers, including the oxygen, carbon, and helium shells, as well as the outer envelope, are left in the precarious position of being almost suspended above the catastrophically collapsing core.

The homologous collapse of the inner core continues until the density there exceeds about $8 \times 10^{17} \text{ kg m}^{-3}$, roughly three times the density of an atomic nucleus. At that point, the nuclear material that now makes up the inner core stiffens because the strong force (usually attractive) suddenly becomes repulsive. This is a consequence of the Pauli exclusion principle applied to neutrons.³ The result is that the inner core rebounds somewhat, sending pressure waves outward into the infalling material from the outer core. When the velocity of the pressure waves reach the sound speed, they build into a shock wave that begins to move outward.

As the shock wave encounters the infalling outer iron core, the high temperatures that result cause further photodisintegration, robbing the shock of much of its energy. For every $0.1 M_{\odot}$ of iron that is broken down into protons and neutrons, the shock loses $1.7 \times 10^{44} \text{ J}$.

Computer simulations indicate that at this point the shock stalls, becoming nearly stationary, with infalling material accreting onto it. In other words, the shock has become an **accretion shock**, somewhat akin to the situation during protostellar collapse, discussed in Section 12.2. However, below the shock, a **neutrinosphere** develops from the processes of photodisintegration and electron capture. Since the overlying material is now so dense that even neutrinos cannot easily penetrate it, some of the neutrino energy ($\sim 5\%$) will be deposited in the matter just behind the shock. This additional energy heats the material and allows the shock to resume its march toward the surface. If this does not happen quickly enough, the initially outflowing material will fall back onto the core, meaning that an explosion doesn't occur.

The success of core-collapse supernova models seems to hinge very sensitively on the details of three-dimensional simulations, which allow hot, rising plumes of gas to mix with colder, infalling gas. The challenges lie in the details of convection, the need to treat the neutrino physics properly (including electron, muon, and tau neutrinos, and their antiparticles), and the very high resolution required for the calculations [up to 10^9 mesh points (locations) in the computational grid]. It may also be necessary ultimately to include a proper treatment of sound waves, differential rotation, and magnetic fields in order to describe all of the observed details of a supernova explosion. This level of computational sophistication challenges even the world's most powerful supercomputers.

Assuming that the scenario just described is essentially correct, and that the shock is able to resume its march to the surface, the shock will drive the envelope and the remainder of the nuclear-processed matter in front of it. The total kinetic energy in the expanding material is on the order of 10^{44} J , roughly 1% of the energy liberated in neutrinos. Finally, when the material becomes optically thin at a radius of about 10^{13} m , or roughly 100 AU, a tremendous optical display results, releasing approximately 10^{42} J of energy in the form of photons, with a peak luminosity of nearly 10^{36} W , or roughly $10^9 L_{\odot}$, which is capable of competing with the brightness of an entire galaxy.

The events just described—the catastrophic collapse of an iron core, the generation of a shock wave, and the ensuing ejection of the star's envelope—are believed to be the general mechanism that creates a core-collapse supernova. The details that result in a Type II rather than a Type Ib or Type Ic supernova have to do with the composition and mass of the envelope at the time of the core collapse and the amount of radioactive material synthesized in the ejecta.

³Neutrons, along with electrons and protons, are fermions.

Type II supernovae, which are more common than either Type Ib or Type Ic, are usually red supergiant stars in the extreme upper-right-hand corner of the H–R diagram at the time they undergo catastrophic core collapse. Type Ib’s and Type Ic’s have lost various amounts of their envelopes prior to detonation. It is now believed that these are the products of exploded Wolf–Rayet stars. Type Ib’s and Type Ic’s may correspond to the detonation of WN and WC Wolf–Rayets, respectively; recall the Conti scenario for the evolution of massive stars described on page 522.

Stellar Remnants of a Core-Collapse Supernova

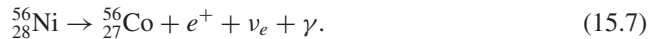
If the initial mass of the star on the main sequence was not too large (perhaps $M_{\text{ZAMS}} < 25 M_{\odot}$), the remnant in the inner core will stabilize and become a **neutron star** (essentially a gigantic nucleus), supported by degenerate neutron pressure.⁴ However, if the initial stellar mass is much larger, even the pressure of neutron degeneracy cannot support the remnant against the pull of gravity, and the final collapse will be complete, producing a **black hole** (an object whose mass has collapsed to a *singularity* of infinite density).⁵ In either case, the creation of these exotic objects is accompanied by a tremendous production of neutrinos, the majority of which escape into space with a total energy on the order of the binding energy of a neutron star, approximately 3×10^{46} J. This represents roughly 100 times more energy than the Sun will produce over its entire main-sequence lifetime!

The Light Curves and the Radioactive Decay of the Ejecta

A Type II-P supernova is the most common type of core-collapse supernova. The source of the plateau in Type II-P light curves is due largely to the energy deposited by the shock into the hydrogen-rich envelope. The gas, which was ionized by the shock, enters a stage of prolonged recombination, releasing the energy at a nearly constant temperature of about 5000 K.

The plateau may be supported further by the energy deposited in the envelope by the **radioactive decay** of ^{56}Ni that was produced by the shock front during its march through the star (the half-life of ^{56}Ni is $\tau_{1/2} = 6.1$ days). It is expected that the explosive nucleosynthesis of the supernova shock should have produced significant amounts of other radioactive isotopes as well, such as ^{57}Co ($\tau_{1/2} = 271$ days), ^{22}Na ($\tau_{1/2} = 2.6$ yr), and ^{44}Ti ($\tau_{1/2} \simeq 47$ yr). If the isotopes are present in sufficient quantities, each in turn may contribute to the overall light curve, causing the slope of the curve to change.

The ^{56}Ni is transformed into ^{56}Co through the *beta-decay* reaction⁶



The energy released by the decay is deposited into the optically thick expanding shell, which is then radiated away from the supernova remnant’s photosphere. This “holds up” the light

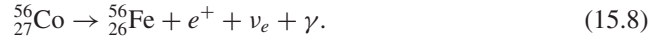
⁴The upper mass limit of the progenitor that results in the formation of a neutron star depends on how metal-rich the original star was. A star sufficiently metal-rich may form a neutron star even if its initial mass is much greater than $25 M_{\odot}$.

⁵We leave the detailed discussion of neutron stars and black holes to Chapters 16 and 17, respectively.

⁶Electrons and positrons are also known as β particles.

curve for a time, extending the observed plateau. Eventually the expanding gas cloud will become optically thin, exposing the central product of the explosion, the neutron star or black hole.

$^{56}_{27}\text{Co}$, the product of the radioactive decay of $^{56}_{28}\text{Ni}$, is itself radioactive, with a longer half-life of 77.7 days:



This implies that as the luminosity of the supernova diminishes over time, it should be possible to detect the contribution to the light being made by $^{56}_{27}\text{Co}$. Type II-L supernovae appear to have had progenitor stars with significantly reduced hydrogen envelopes, implying that the signature of the radioactive decay becomes evident almost immediately after the event.

Since radioactive decay is a statistical process, the rate of decay must be proportional to the number of atoms remaining in the sample, or

$$\frac{dN}{dt} = -\lambda N, \quad (15.9)$$

where λ is a constant. It is left as an exercise to show that Eq. (15.9) can be integrated to give

$$N(t) = N_0 e^{-\lambda t}, \quad (15.10)$$

where N_0 is the original number of radioactive atoms in the sample (see Fig. 15.11), and

$$\lambda = \frac{\ln 2}{\tau_{1/2}}.$$

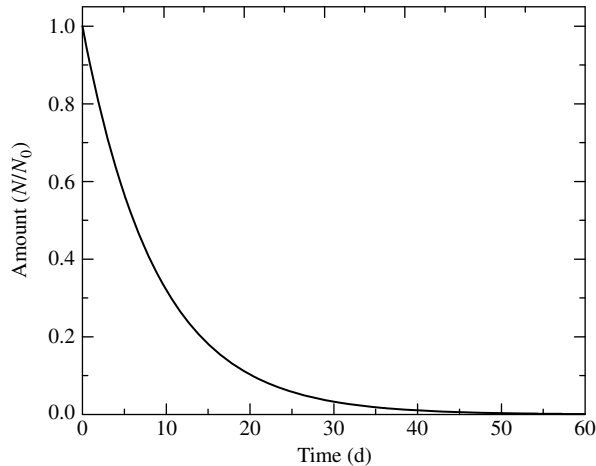


FIGURE 15.11 The radioactive decay of $^{56}_{28}\text{Ni}$, with a half-life of $\tau_{1/2} = 6.1$ days. There is a 50% chance that any given $^{56}_{28}\text{Ni}$ atom will decay during a time interval of 6.1 days. If the original sample is entirely composed of $^{56}_{28}\text{Ni}$, after n successive half-lives the fraction of Ni atoms remaining is 2^{-n} .

Since the rate at which decay energy is being deposited into the supernova remnant must be proportional to dN/dt , the slope of the bolometric light curve is given by

$$\frac{d \log_{10} L}{dt} = -0.434\lambda \quad (15.11)$$

or

$$\frac{dM_{\text{bol}}}{dt} = 1.086\lambda. \quad (15.12)$$

Therefore, by measuring the slope of the light curve, we can determine λ and verify the presence of large quantities of a specific radioactive isotope, like ^{56}Co .

Given its proximity to Earth, the most carefully studied supernova to date has been SN 1987A. However, almost as soon as it was discovered, astronomers realized that SN 1987A was unusual when compared with other, more distant Type II's that had been observed. This was most evident in the rather slow rise to maximum light (taking 80 days), which peaked only at an absolute bolometric magnitude of -15.5 , whereas a typical Type II reaches $M_{\text{bol}} = -18$. The light curve through day 1444 after the outburst is shown in Fig. 15.12.

The decay of the $0.075 M_{\odot}$ of ^{56}Ni that was produced by the shock occurred while the timescale required for energy to be radiated away was still quite long. Consequently, the

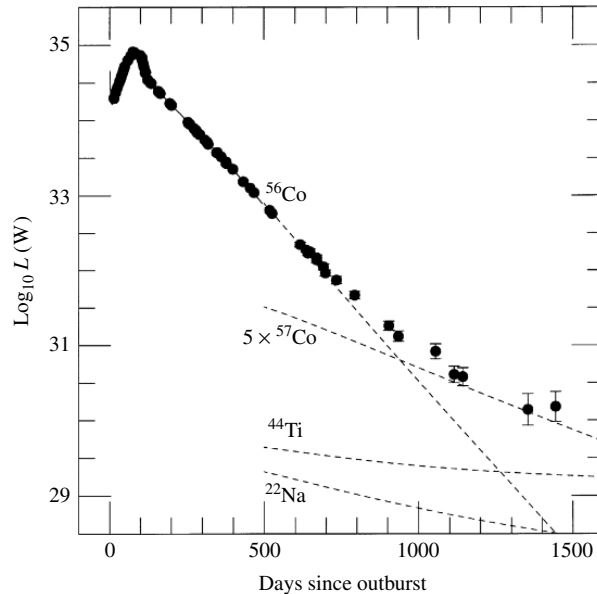


FIGURE 15.12 The bolometric light curve of SN 1987A through the first 1444 days after the explosion. The dashed lines show the contributions expected from the radioactive isotopes produced by the shock wave. The initial masses are estimated to be ^{56}Ni (and later ^{56}Co), $0.075 M_{\odot}$; ^{57}Co , $0.009 M_{\odot}$ (five times the solar abundance); ^{44}Ti , $1 \times 10^{-4} M_{\odot}$; and ^{22}Na , $2 \times 10^{-6} M_{\odot}$. (Figure adapted from Suntzeff et al., *Ap. J. Lett.*, 384, L33, 1992.)

added decay energy produced a bump on the light curve near maximum light, rather than forming a plateau. By the time the resulting $^{56}_{27}\text{Co}$ began to decay, this diffusion timescale had become sufficiently short that the decrease in the luminosity of the remnant began to track closely the rate of decay of cobalt-56. Subsequently, the next important radioactive isotope, $^{57}_{27}\text{Co}$, began to play an important role in the development of the light curve. The expected contributions of the various radioactive isotopes to the light curve of SN 1987A are shown in Fig. 15.12, the slopes in the light curve being related to the half-lives of the isotopes through Eq. (15.11).

SN 1987A also allowed astronomers, for the first time, to directly measure the X-ray and gamma-ray emission lines produced by radioactive decay. In particular, the 847-keV and 1238-keV lines of $^{56}_{27}\text{Co}$ were detected by a number of experiments, confirming the presence of this isotope. Doppler shift measurements indicate that the heavier isotopes in the remnant are expanding at several thousand kilometers per second.

The Subluminous Nature of SN 1987A

The mystery of the subluminous nature of SN 1987A was solved when the identity of its progenitor was established. The star that blew up was the 12th-magnitude *blue* supergiant (spectral class B3 I), Sk −69 202. Since what exploded was a much smaller blue supergiant, rather than a red supergiant (as is usually assumed to be the case), the star was more dense. As a result, before the thermal energy produced by the shock could diffuse out and escape as light, it was converted into the mechanical energy required to lift the envelope of the star out of the deeper potential well of a blue supergiant. Measurements of H α lines indicate that some of the outer hydrogen envelope was ejected at speeds near 30,000 km s^{−1}, or 0.1c!

The available observations of Sk −69 202, together with theoretical evolutionary models, suggest that the progenitor of SN 1987A had a mass of roughly 20 $M_{\odot}$ when it was on the main sequence and that it lost perhaps a few solar masses before its iron core collapsed (estimated to be between 1.4 and 1.6 $M_{\odot}$). Although it was apparently a red supergiant for between several hundred thousand and one million years, it evolved to the blue just 40,000 years before the explosion (refer to the evolutionary tracks in Fig. 15.3; the blueward loop of a 20 $M_{\odot}$ star is not shown, but the blueward loop of a 25 $M_{\odot}$ star is present on the diagram.). Supporting this hypothesis is the observation that hydrogen was more abundant in the envelope of Sk −69 202 than was helium, suggesting that the star had not suffered extensive amounts of mass loss. Whether and when a massive star evolves from being a red supergiant to a blue supergiant before exploding depends sensitively on the mass of the star (it cannot be much more than about 20 $M_{\odot}$), its composition (it must be metal poor, as are the stars of the LMC), the rate of mass loss (which must be low), and the treatment of convection (always a major uncertainty in theoretical stellar models).

Supernova Remnants

There are now many examples of **supernova remnants** (SNR), including the Crab Nebula, located in the constellation of Taurus (recall Fig. 15.4). Today, nearly 1000 years since the SN 1054 explosion, the Crab is still expanding at a rate of almost 1450 km s^{−1} and it has a luminosity of $8 \times 10^4 L_{\odot}$. Much of the radiation being emitted is in the form of highly polarized synchrotron radiation (see Section 4.3), indicating the presence of relativistic

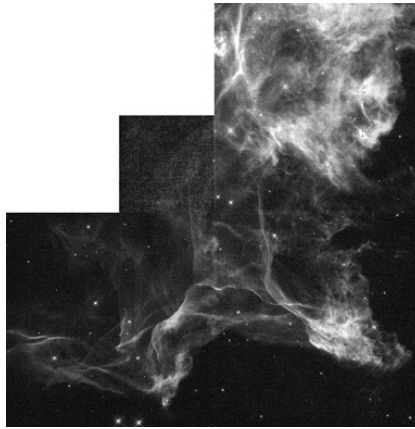


FIGURE 15.13 An HST WF/PC 2 image of a portion of the Cygnus Loop, 800 pc away. (Courtesy of J. Hester/Arizona State University and NASA.)

electrons that are spiraling around magnetic field lines. The ongoing source of the electrons and the continued high luminosity so long after the explosion remained major puzzles in astronomy until the discovery of a **pulsar** (a rapidly spinning neutron star) at the center of the Crab SNR. Pulsars will be discussed extensively in Chapter 16.

A second example of a supernova remnant is shown in Fig. 15.13. The image is of a small portion of the 15,000-year-old Cygnus Loop nebula, located 800 pc from Earth in the constellation of Cygnus. The remnant is expanding from left to right in the image, producing shock fronts several astronomical units wide as the debris from the supernova explosion encounters material in the interstellar medium. The shocks excite and ionize the ISM, causing the observed emission.

Although mass loss prior to the explosion could not have been excessive, the progenitor of SN 1987A did lose some mass, resulting in a very unusual structure around the expanding supernova remnant. The Hubble Space Telescope has recorded three rings around SN 1987A (Fig. 15.14). The innermost ring measures 0.42 pc in diameter and lies in a plane that contains the center of the supernova explosion. It glows in visible light as a consequence of emissions from O III energized by radiation from the supernova and appears elongated because it is inclined relative to our line of sight. The material making up the central ring was ejected by stellar winds 20,000 years before the explosion of SN 1987A.

The two larger rings are not in planes containing the central explosion but lie in front of and behind the star. One explanation for these fascinating and unexpected features is that Sk −69 202 resided near a companion star, possibly a neutron star or a black hole (see Chapters 16 and 17, respectively). As this companion source wobbles, narrow jets of radiation from the source “paint” the rings on an hourglass-shaped, bipolar distribution of mass that was ejected from Sk −69 202. It is in the denser equatorial plane of the bipolar mass distribution that the central ring is located. In support of this hypothesis, researchers believe that they may have identified the source of these beams of radiation about 0.1 pc from the center of the supernova explosion, consistent with the fact that the larger rings appear to be offset from the explosion’s center. Opponents of this model believe that the explanation is too complicated: It requires two sources of high-energy radiation, one to

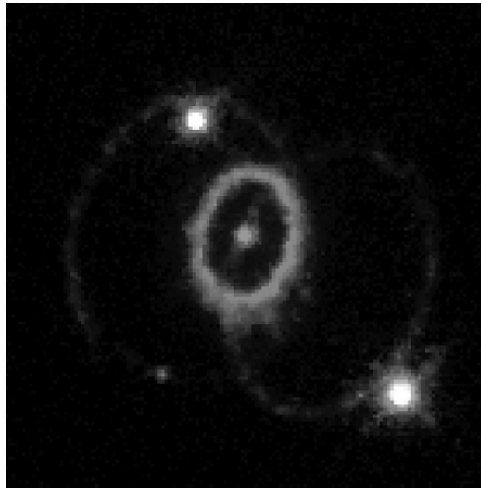


FIGURE 15.14 Rings around SN 1987A, detected by the Hubble Space Telescope in 1994. The diameter of the inner ring is 0.42 pc. (Courtesy of Dr. Christopher Burrows, ESA/STScI and NASA.)

explain the central ring and another to explain the larger ones. Alternatively, the larger rings could be the product of a hot, fast stellar wind from the blue supergiant progenitor overtaking the slower, cooler wind given off by the star when it was still a red supergiant.

In the summer of 1990, fluctuating radio emissions were finally detected from the supernova. Although radio wavelength energy was detected during the first days following the explosion, SN 1987A had remained radio quiet since that time. Apparently the shock wave, still propagating outward at a speed close to $0.1c$, collided with clumps of material lost from Sk −69 202 prior to the supernova event.

The shock front from the expanding supernova remnant of SN 1987A began to collide with the slower-moving stellar wind comprising the inner ring in 1996. The result was a brilliant display of bright clumps in the inner ring that developed over the next several years. The expanding shock front and the inner ring are shown in successive images in Fig. 15.15.

The Detection of Neutrinos from SN 1987A

Arguably the most exciting early observations of SN 1987A were based on its neutrinos, representing the first time that neutrinos had been detected from an astronomical source other than the Sun. The measurement of the neutrino burst confirmed the basic theory of core-collapse supernovae and amounts to our “seeing” the formation of a neutron star out of the collapsed iron core.

The arrival of the neutrino burst was recorded over a period of $12\frac{1}{2}$ seconds, beginning at February 23.316 UT, 1987, three hours *before* the arrival of the photons at February 23.443 UT. Twelve events were recorded at Japan’s Kamiokande II Cerenkov detector, and at the same time, eight events were detected by the underground IMB⁷ Cerenkov detector near

⁷IMB stands for the consortium that operates the observatory: University of California at Irvine, University of Michigan, and Brookhaven National Laboratory.

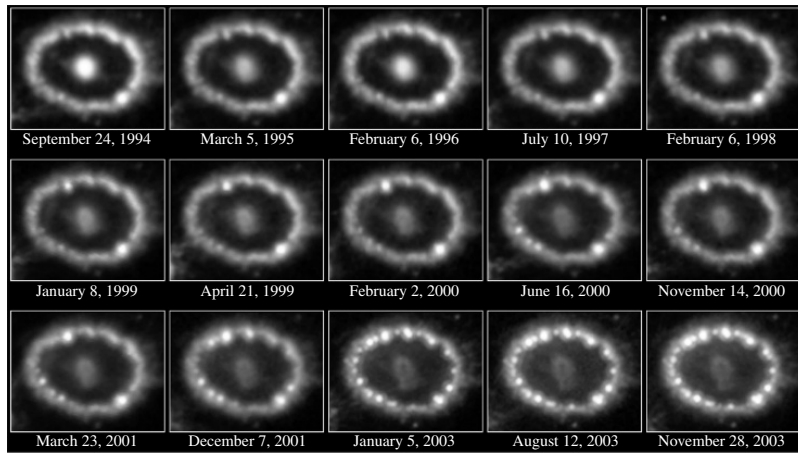


FIGURE 15.15 The expanding nebula from SN 1987A is seen in the center of this series of images. The inner ring shown in Fig. 15.14 is being overtaken by the shock front, causing clumps of gas to glow. The bright spot visible in the lower-right-hand portion of the ring in every image is another star that happens to be situated in the line of sight of the ring. [NASA and R. Kirshner (Harvard–Smithsonian Center for Astrophysics).]

Fairport, Ohio.⁸ Assuming that the exploding star became optically thin to neutrinos before the shock wave reached the surface, and assuming further that the neutrinos traveled faster than the shock while still inside the star, the neutrinos began their trip to Earth ahead of the photons. Given the fact that the neutrinos arrived ahead of the light, their velocity through space must have been very near the speed of light (within one part in 10^8). This observation, together with the absence of any significant dispersion in the arrival time of neutrinos of different energies (i.e., higher-energy neutrinos did not arrive any earlier than lower-energy ones), suggests that the rest mass of electron neutrinos must be quite small. The upper limit on the electron neutrino, based on data from SN 1987A, is $m_e \leq 16$ eV, consistent with the results of laboratory experiments that place the upper limit at 2.2 eV.

The Search for a Compact Remnant of SN 1987A

Interestingly, as of 2006, neutrinos have been the only direct evidence of the formation of a compact object at the center of SN 1987A. All attempts to detect a remnant in optical, ultraviolet, or X-ray wavelengths have failed. In addition, efforts to find any evidence of a surviving binary companion have also been unsuccessful. The upper limit on the luminosity in the optical portion of the electromagnetic spectrum is currently less than 8×10^{26} W, equivalent to the optical energy output of an F6 main-sequence star. Ultraviolet spectra lead to an upper limit of $L_{UV} \leq 1.7 \times 10^{27}$ W, and Chandra has set an upper limit on the X-ray luminosity of $L_X \leq 5.5 \times 10^{26}$ W in the energy band between 2 and 10 keV.

⁸The famous Davis solar neutrino detector (discussed extensively in Section 11.1) did not measure any neutrinos from SN 1987A; the solar neutrino background was much *larger* than the neutrino count from the supernova in the energy range of the detector!

These limits are sufficiently stringent that models for the form of the compact companion and its environment (thin accretion disk, thick accretion disk, spherical accretion, etc.) are becoming seriously restricted. Perhaps future infrared observations with Spitzer will finally detect the elusive compact remnant.

Although SN 1987A has presented some interesting twists in our study of stellar evolution, it has also confirmed or clarified important aspects of the theory.

Chemical Abundance Ratios in the Universe

Following the discussion of core-collapse supernovae, and in anticipation of results from our discussion of Type Ia supernovae coming up in Section 18.5, it is worth revisiting the chemical compositions and observed abundance ratios in the universe. A critical component in determining the success of current stellar evolution theory is the ability to explain the observed abundance ratios of the elements.

The chemical composition of the Sun's photosphere is shown in Fig. 15.16, with all values normalized to 10^{12} for hydrogen; see also Table 9.2. By far the most abundant element in the universe is hydrogen, with helium being less abundant by about a factor of 10. It is believed that hydrogen is primordial, having been synthesized immediately following the Big Bang that began the universe. Much of the present-day helium was also produced directly from the Big Bang, while the remainder was generated from hydrogen burning in stellar interiors.

Relative to hydrogen and helium, lithium, beryllium, and boron are very under-abundant. There are two reasons for this: They are not prominent end products of nuclear reaction chains, and they can be destroyed by collisions with protons. For lithium this occurs at

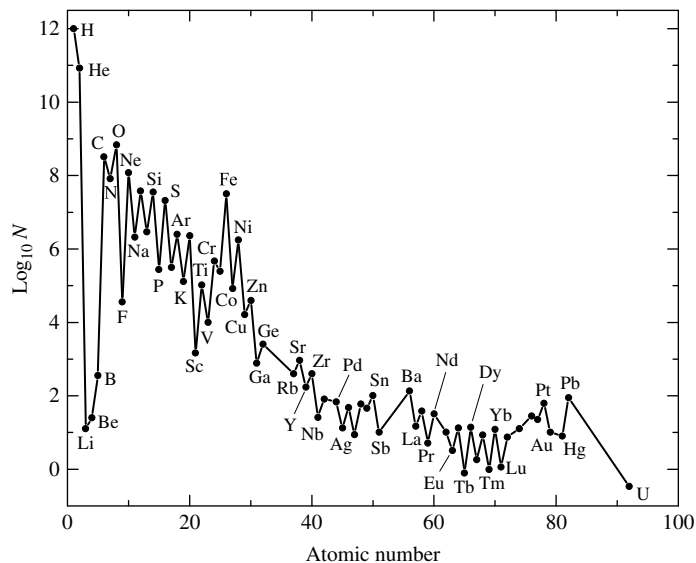


FIGURE 15.16 The relative abundances of elements in the Sun's photosphere. All abundances are normalized relative to 10^{12} hydrogen atoms. (Data from Grevesse and Sauval, *Space Sci. Rev.*, 85, 161, 1998.)

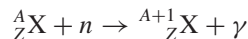
temperatures greater than about 2.7×10^6 K, while for beryllium the required temperature is 3.5×10^6 K. It is the Sun's surface convection zone that is responsible for transporting the surface lithium, beryllium, and boron into the interior. When the present-day solar composition is compared with the abundances of meteorites (which should be similar to the Sun's primordial composition), we find that the relative abundances of beryllium are comparable but that the Sun's surface composition of lithium is smaller than the meteorites' lithium abundance by a factor of about 100. This suggests that lithium has been destroyed in the Sun since the star's formation but that beryllium has not been appreciably depleted. Apparently the base of the solar convection zone extends down sufficiently far to burn lithium but not far enough to burn beryllium. However, combining stellar structure theory, including the mixing-length theory of convection, with the analysis of solar oscillations (see Section 14.5) indicates that the base of the convection zone extends down to 2.3×10^6 K, not far enough to burn lithium adequately. The disagreement of standard models with the observations is known as the **solar lithium problem**.⁹

Peaks occur in Fig. 15.16 for elements such as carbon, nitrogen, oxygen, neon, and so on because they are created as a consequence of a star's evolutionary trek toward the iron peak and because they are relatively stable α -particle-rich nuclei (recall Fig. 10.9).

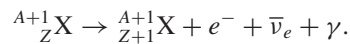
Core-collapse supernovae are also responsible for the generation of significant quantities of oxygen, and Type Ia supernovae are responsible for the creation of most of the iron observed in the cosmos.

s-Process and r-Process Nucleosynthesis

When nuclei having progressively higher values of Z (the number of protons) form via stellar nucleosynthesis, it becomes increasingly difficult for other charged particles, such as protons, alpha particles, and so on, to react with them. The cause is the existence of a high Coulomb potential barrier. However, the same limitation does not exist when neutrons collide with these nuclei. Consequently, nuclear reactions involving neutrons can occur even at relatively low temperatures, assuming, of course, that free neutrons are present in the gas. The reactions with neutrons



result in more massive nuclei that are either stable or unstable against the beta-decay reaction,



If the beta-decay half-life is short compared to the timescale for neutron capture, the neutron-capture reaction is said to be a *slow process* or an **s-process** reaction (recall the discussion on page 467). s-Process reactions tend to yield stable nuclei, either directly or secondarily via beta decay. On the other hand, if the half-life for the beta-decay reaction is long compared with the time scale for neutron capture, the neutron-capture reaction is termed a *rapid*

⁹Perhaps because of their momenta, descending convective bubbles *overshoot* the bottom of the zone of convective instability, causing lithium to be transported deeper than the standard models suggest. Additional effects may also derive from diffusion and the interaction of convection with rotation.

process or *r-process* and results in neutron-rich nuclei. *s*-Process reactions tend to occur in normal phases of stellar evolution, whereas *r*-processes can occur during a supernova when a large flux of neutrinos exists. Although neither process plays significant roles in energy production, they do account for the abundance ratios of nuclei with $A > 60$.

15.4 ■ GAMMA-RAY BURSTS

One of the many great detective stories of modern astrophysics began in the 1960s with the launch of the Vela series of military satellites. The Vela spacecraft were designed to monitor compliance of the former Soviet Union with the 1963 nuclear test ban treaty by looking for sudden bursts of gamma rays of terrestrial origin. By 1967 it was clear that the **gamma-ray bursts** (GRB) that were being detected were coming from above rather than below, but it was not until 1973 that this information was released to the public.

About once per day, at some random location in the sky, a shower of gamma-ray photons with energies ranging from about 1 keV to many GeV appears. (Although the lower end of this range includes X-ray photons, most of the energy is in gamma rays.) The bursts last from 10^{-2} to 10^3 s, and they have rise times as fast as 10^{-4} s, followed by an exponential decay. The bursts are usually multiply peaked and complex, although there is no typical burst profile. Two examples of gamma-ray bursts recorded by the Burst and Transient Source Experiment (BATSE) onboard the Compton Gamma-Ray Observatory (CGRO) are shown in Fig. 15.17.

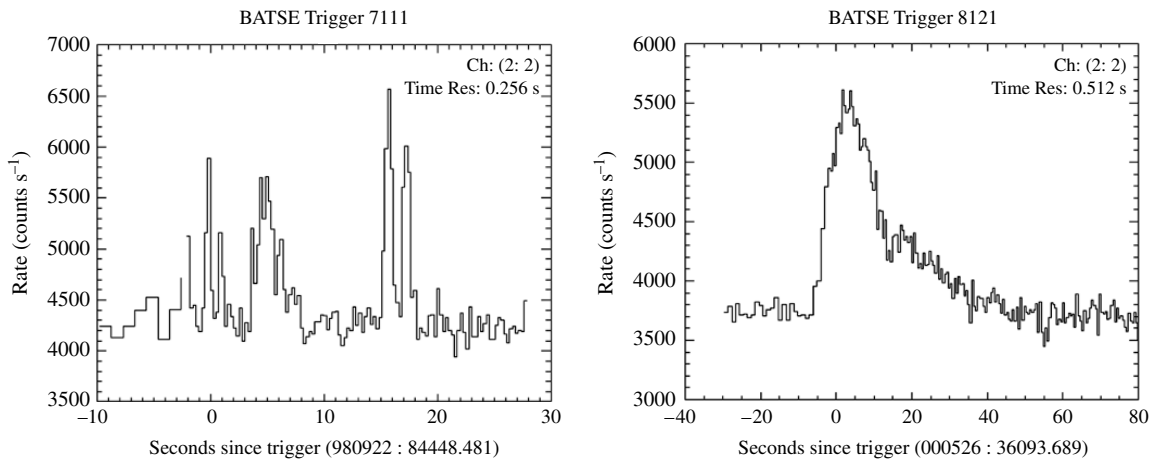


FIGURE 15.17 Light curves of two gamma-ray bursts, GRB 980922 and GRB 000526, in the energy range between 50 keV and 100 keV. The data were obtained by BATSE onboard the Compton Gamma-Ray Observatory. The dates of the two events are recorded in their designations; GRB 980922 occurred on September 22, 1998, and GRB 000526 occurred on May 26, 2000. GRB 000526 was the last gamma-ray burst recorded by BATSE before the Compton Gamma-Ray Observatory was deorbited. (Courtesy of the BATSE Team – NASA.)

Are the Sources of GRBs Galactic or Extragalactic?

Until the late 1990s much of the mystery of GRBs was associated with their distances and the resulting implied energy of the events. It was not clear whether gamma-ray bursts originated in our Solar System (perhaps in the Oort cloud of comets), within our Galaxy, or the farthest reaches of the universe.

Without knowing the distance to the sources of the gamma rays, we must give their energy output in terms of the total energy received per unit area of detector surface during the burst (the energy flux integrated over the duration of the burst). This quantity, called the **fluence**, S , can be as small as $10^{-12} \text{ J m}^{-2}$ or as great as 10^{-7} J m^{-2} . The most energetic bursts recorded by the CGRO saturated its detectors. For example, the “Super Bowl burst,” named because it occurred on Super Bowl Sunday (January 31) in 1993, lasted only a second, but an afterglow of energetic photons (up to 1 GeV each) persisted for about 100 seconds. Then, on December 15, 1994, the CGRO measured a burst that lasted for 90 minutes, with a peak photon energy of 18 GeV!

Example 15.4.1. Suppose that the fluence of a particular GRB was determined to be 10^{-7} J m^{-2} . Assuming that the source of the burst was located 50,000 AU away in the Oort cloud of comets within our Solar System, and assuming further that the emission of the energy from the source was isotropic, the energy of the burst would necessarily have been

$$E = (4\pi r^2) S = 4\pi (50,000 \text{ AU})^2 (10^{-7} \text{ J m}^{-2}) = 7 \times 10^{25} \text{ J}.$$

On the other hand, if the source of the GRB were located 1 Gpc away in a distant galaxy, then the amount of energy involved in the burst (again assuming isotropy) would have been

$$E = (4\pi r^2) S = 4\pi (1 \text{ Gpc})^2 (10^{-7} \text{ J m}^{-2}) = 1 \times 10^{45} \text{ J},$$

comparable to the energy released in a Type II supernova, including its neutrino emission.

The difference between the two estimates is almost 20 orders of magnitude! Clearly it is critical to understand the distance to these objects in order to begin to understand the underlying process.

Even before the CGRO was released by the Space Shuttle *Atlantis* on April 5, 1991, most astronomers agreed that the gamma-ray burst mechanism involved the presence of a neutron star. The short rise times of the bursts, when multiplied by the speed of light, result in a characteristic length comparable to the size of a neutron star (ct_{rise} as small as 30 km). Observations of emission lines corresponding to photons with energies of roughly 350 to 500 keV are thought to be due to the 511-keV photons that are created when an electron and a positron annihilate each other and produce two gamma-ray photons near the surface of a neutron star. The energy of the photons is reduced by as much as 25% as they climb out of the severe potential well of the neutron star. Other spectral features with photons in the range of 20–60 keV have been identified as cyclotron lines due to a 10^8 T magnetic field, typical of the fields of pulsars. The consensus was that the gamma-ray bursts were produced by neutron stars in the thick disk of the Galaxy at distances of a few hundred parsecs; the GRBs occurred either by tapping their internal energy (as in a pulsar glitch) or through accretion in a close binary system.

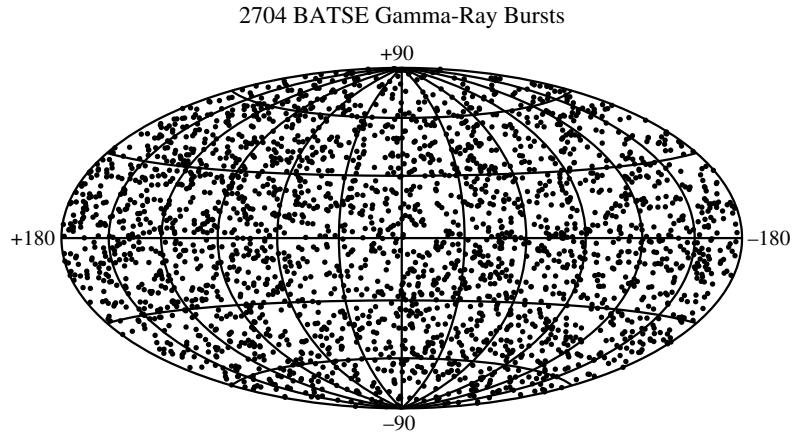


FIGURE 15.18 The isotropic angular distribution of 2704 gamma-ray bursts seen by the BATSE detector onboard the CGRO. (Courtesy of the BATSE Team – NASA.)

There were problems with this scenario. If the bursts were due to accretion in a binary system, why didn't they repeat? Furthermore, the bursts were distributed uniformly over the celestial sphere, instead of being concentrated in the plane of the Milky Way like most pulsars and X-ray binaries. It was thought that this could be due to the relative insensitivity of the gamma-ray detectors employed in the pre-Compton era. These instruments were unable to see sources beyond the disk of stars in our part of the Galaxy, and so the distribution of bursts appeared to be isotropic. (In the same manner, if we could see only the nearest stars, they would be uniformly scattered over the night sky instead of being concentrated along the Milky Way.) It was thought that this situation would surely change with the launch of the much more sensitive Compton Gamma Ray Observatory satellite. On average, one burst was observed every 25 hours by the spacecraft. Figure 15.18 shows the distribution of 2704 gamma-ray bursts observed by BATSE; there is no statistically significant deviation from an isotropic distribution.

Equally interesting was the finding that although the sources are spread evenly across the sky, they do not appear to be distributed homogeneously throughout space.¹⁰ A classic argument was employed to determine whether there is an edge to the distribution. Let E be the energy of a gamma-ray burst, located at a distance r from Earth. Then the fluence is

$$S = \frac{E}{4\pi r^2}, \quad (15.13)$$

assuming an isotropic burst. Solving this expression for r , we have

$$r(S) = \left(\frac{E}{4\pi S} \right)^{1/2}.$$

¹⁰A homogeneous distribution would have the same number density of sources everywhere, independent of distance or direction.

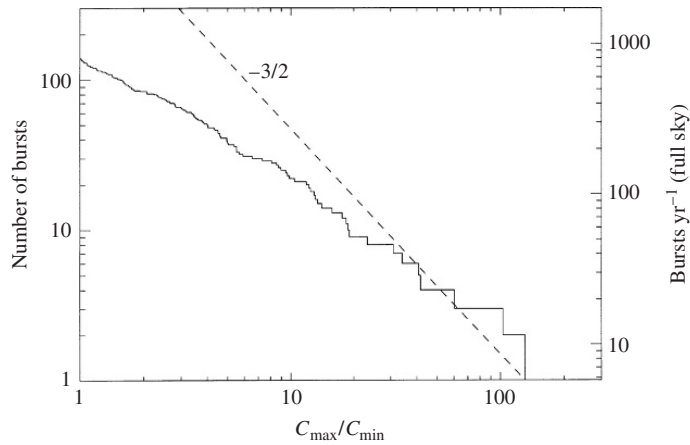


FIGURE 15.19 Violation of the proportionality $N \propto S^{-3/2}$, indicating an edge to the distribution of gamma-ray burst sources. The maximum gamma-ray count rate, $C_{\max}$, is plotted instead of the fluence, S ; $C_{\min}$ is the weakest the CGRO can confidently detect. (Figure adapted from Meegan et al., *Nature*, 355, 143, 1992.)

Assuming that all burst sources have the same intrinsic energy, E , then for a specific value of S (say, S_0), all of the sources within a sphere of radius $r(S_0)$ will be observed to have a fluence $S \geq S_0$.¹¹ If there are n burst sources per unit volume, then the number of sources with a fluence equal to or greater than S_0 is

$$N(S) = \frac{4}{3}\pi n r^3(S) = \frac{4}{3}\pi n \left(\frac{E}{4\pi S} \right)^{3/2}, \quad (15.14)$$

where the “0” subscript has been dropped. Thus, if the sources of the gamma-ray bursts are distributed uniformly throughout space, the number of bursts observed with a fluence greater than some value S would be proportional to $S^{-3/2}$. The Compton results show that this proportionality is violated when S is small enough to include the more distant, fainter, sources; see Fig. 15.19.¹² This implies that there is an edge to the distribution; the burst sources do not extend outward without limit (but of course the limit could be the edge of the observable universe). When this result is combined with the fact that the distribution is isotropic, the implication is that Earth is near the center of a spherically symmetric distribution of gamma-ray burst sources.

The resolution to the distance question finally came with the detection of GRB 970228 on February 28, 1997, by the BeppoSAX spacecraft. (BeppoSAX is the product of a collaboration between the Italian Space Agency and the Netherlands Agency for Aerospace Programs.) The Gamma-Ray Burst Monitor on BeppoSAX first noticed the event, allowing

¹¹ It may be that there are different populations of burst sources with different characteristic values of E . However, the following argument is still valid if each population is homogeneously distributed.

¹² Note that in Fig. 15.19, the maximum count rate is plotted instead of the fluence. It is the maximum count rate that determines whether a burst is detected, making it more appropriate to use in the statistics of burst counts.

its wide-field X-ray camera to localize the region of the sky to within 3'. Within a few hours, the observatory's narrow-field X-ray telescopes were able to further localize the source. With rapid knowledge of the position of the GRB, it became possible for other ground-based and orbiting observatories to investigate that area of the sky as well. Even after the gamma-ray signature had vanished, fading X-ray and optical counterparts were detected. Deep images of the region, obtained using the Keck Observatory and the Hubble Space Telescope, revealed that the GRB had occurred in a distant galaxy, indicating that GRB 970228 originated at a cosmological (extragalactic) distance.

Additional cosmological discoveries of GRBs from BeppoSAX and other rapid-response gamma-ray telescopes [including NASA's High Energy Transient Explorer (HETE-2) and Swift missions] were quickly followed up with optical identifications that have verified the cosmological distances involved in GRB events. Numerous GRBs have since been associated with fading X-ray, optical, and radio counterparts in distant galaxies. This implies that GRBs are among the most energetic phenomena in the universe, comparable to the staggering energy release of core-collapse supernovae.

Two Classes of GRBs

Now that the true distances to gamma-ray bursts can be determined and the scale of the energy output confirmed, it becomes possible to evaluate proposed mechanisms for generating GRBs. After the study of thousands of events, it is clear that there are two basic classes of gamma-ray bursts. Those events that last longer than 2 seconds are referred to as **long-soft** GRBs, while those that are shorter than 2 seconds are **short-hard** events. "Soft" and "hard" refer to having more of the event energy at lower energies or higher energies, respectively.

Just as there are two fundamentally different types of supernovae (Type Ia and core-collapse), it appears that there are two fundamentally different types of gamma-ray bursts. The short-hard bursts seem to be associated with neutron star–neutron star or neutron star–black hole mergers, whereas long-soft bursts may be connected with supernovae. As with Type Ia supernovae, we will defer the discussion of short-hard GRBs until later (Section 18.6), after we have had an opportunity to study the physics of compact objects more thoroughly.

Core-Collapse Supernovae and Long-Soft GRBs

A direct link between a supernova and a long-soft GRB was established with the detection of GRB 980425. At a distance of 40 Mpc, GRB 980425 was determined to be about five orders of magnitude less energetic (8×10^{40} J) than a typical GRB. What was particularly important, however, was the detection of a supernova, SN 1998bw, at the same location. It seems that SN 1998bw was a particularly energetic Type Ib or Ic supernova with a total energy output of between 2 and 6×10^{45} J (about 30 times greater than a typical Type Ib/c). It is likely that the remnant core that collapsed was $3 M_{\odot}$, resulting in a black hole.

A second GRB–supernova connection has also been identified, this one between GRB 030329 and SN 2003dh. In this case the energy of the gamma-ray burst is more typical of other GRBs.

Models of Long–Soft GRBs

Several models have been proposed to explain long–soft gamma-ray bursts, but one common ingredient in the models involves the *beaming of highly relativistic matter*. The prodigious amount of energy that is apparently produced in a GRB may be reduced by replacing the assumption of isotropic emission (recall Example 15.4.1) with **relativistic jets**, illustrated schematically in Fig. 15.20. According to Eq. (4.43), radiation is emitted in a cone having an opening angle of half-width $\theta \sim 1/\gamma$ for $\gamma \gg 1$ (see Fig. 4.10), where

$$\gamma \equiv \frac{1}{\sqrt{1 - u^2/c^2}}$$

is the Lorentz factor (see Eq. 4.20). If the emitted radiation is beamed forward (rather than isotropically) as the jet of material advances at speeds near the speed of light, and if Earth lies within the beam, the amount of energy produced by the GRB will seem to be greater than it actually is. Models suggest that Lorentz factors of up to 100 or more may be possible in gamma-ray bursts, implying that the actual energy produced and emitted into the solid angle of the jet may be smaller by a factor of γ^2 —perhaps 10,000 times less than the isotropic assumption would suggest.

One challenge with the concept of a highly relativistic jet is that expanding material could encounter baryon-rich material, causing the jet to slow down. This would happen after the jet material had swept up matter with a total rest-mass energy on the order of γmc^2 , where m is the mass of the jet’s material.

The first viable model proposed to explain long–soft bursts is the **collapsar** model of Stan Woosley (sometimes also referred to as a *hypernova* model). As we will learn in Section 16.6, neutron stars have an upper mass limit based on the ability of neutron degeneracy to support the extremely compact star (similar to electron degeneracy pressure). Using sophisticated equations of state for neutrons at very high densities ($\rho \sim 10^{18} \text{ kg m}^{-3}$), investigators have estimated that the maximum mass of a nonrotating neutron star is about $2.2 M_\odot$. When a core-collapse supernova occurs, either a neutron star or a black hole will form, depending on the mass, metallicity, and rotation of the progenitor star. Woosley’s models have suggested that for a progenitor star with sufficiently great mass (possibly a Wolf–Rayet star), the central object to form will be a black hole with a debris disk surrounding it. The collimating effect of the debris disk and associated magnetic fields would lead to a jet emanating from the center of the supernova. Since the jet material will be highly relativistic, it will appear to be further collimated. The jet will plow its way through the overlying material of the infalling stellar envelope producing bursts of gamma-rays. One version of Woosley’s collapsar model is shown in Fig. 15.21.

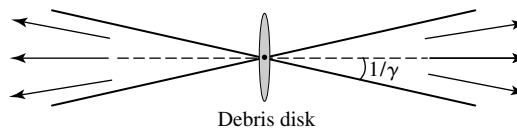


FIGURE 15.20 A relativistic jet of material will appear to the observer as having a cone opening of half-width $\theta \sim 1/\gamma$.

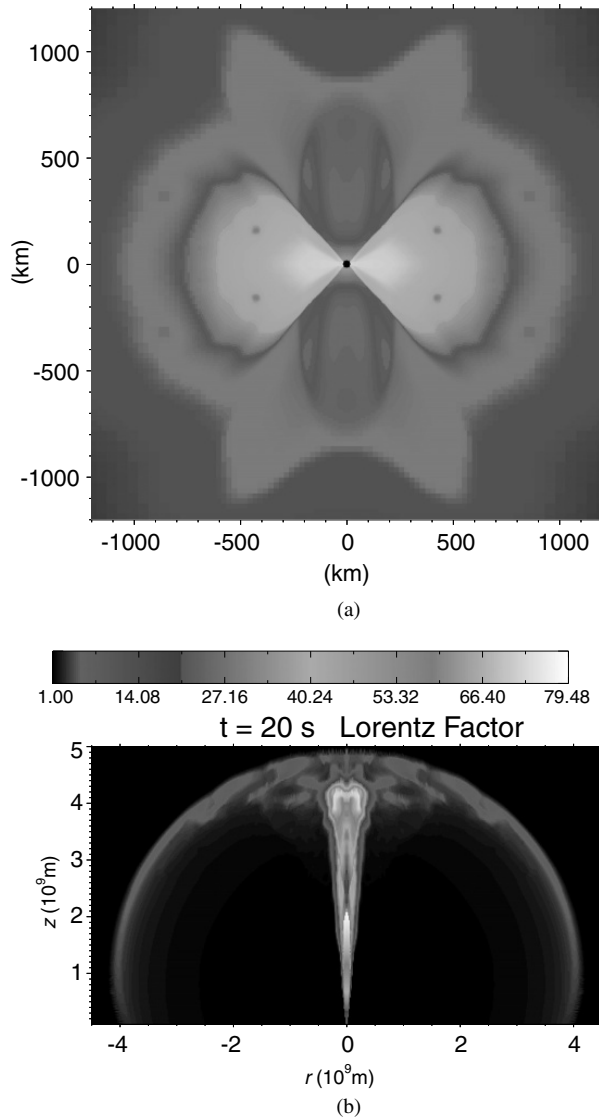


FIGURE 15.21 A collapsar model of the formation of a gamma-ray burst event. (a) The central region of the star following the formation of a black hole and debris disk. (b) The emerging relativistic jet. (Figures courtesy of Weiqun Zhang and Stan Woosley.)

An alternative to the collapsar model is the **supranova** model. Rather than the black hole forming immediately during the core-collapse supernova, a delay in the mechanism occurs. Although the upper mass limit of a static neutron star is believed to be $2.2 M_{\odot}$, a rapidly rotating neutron star can be supported with a mass of up to $2.9 M_{\odot}$. Neutron stars are also likely to have very strong magnetic fields associated with them that could lead to a slowing of the rotation rate.

The supranova model suggests that a supermassive rotating neutron star may form with $M > 2.2 M_{\odot}$ from a core-collapse supernova that would then slow down over the course of weeks or months until it was no longer stable against further gravitational collapse. A catastrophic collapse to a black hole would then result. If a debris disk were to form around the black hole, the disk, in combination with the magnetic field, could produce a relativistic jet and a gamma-ray burst. One advantage of this model is that it would naturally explain the absence of baryons that would slow the jet since the envelope of the star would have been swept up in the supernova explosion that happened earlier.

15.5 ■ COSMIC RAYS

On August 7, 1912, Victor F. Hess (1883–1964) and two colleagues ascended in a balloon to an altitude of 5 km. During that six-hour flight, Hess made careful readings from three electroscopes that he used to measure the intensity of radiation.¹³ As the balloon rose, Hess determined that the level of radiation increased with altitude. From his experiment he concluded, “The results of these observations seem best explained by a radiation of great penetrating power entering our atmosphere from above...” This event marked the birth of the study of **cosmic rays**.¹⁴

Charged Particles from Space

Although referred to as “rays,” this penetrating radiation is actually composed of charged particles. A wide range of masses and classes of particles have been identified in cosmic rays, from electrons, positrons, protons, and muons to a host of nuclei including, but not limited to, carbon, oxygen, neon, magnesium, silicon, iron, and nickel (products of stellar nucleosynthesis).

Particularly striking is the wide range of energies involved, from less than 10^7 eV to at least 3×10^{20} eV. Near the low end of the energy spectrum, cosmic rays impinge on the atmosphere with fluxes of more than $1 \text{ particle m}^{-2} \text{ s}^{-1}$, whereas the highest-energy cosmic rays have fluxes that are very low: less than $1 \text{ particle km}^{-2} \text{ century}^{-1}$. The flux as a function of energy is shown in Fig. 15.22.

Sources of Cosmic Rays

The question naturally arises as to what the source (or sources) of these particles may be. One obvious answer is our Sun. As we saw in Chapter 11, the solar wind, flares, and coronal mass ejections emit charged particles into space on a routine basis. Although abundant, these **solar cosmic rays** (also known as **solar energetic particles**) are relatively low-energy particles as cosmic rays go. Recall that the fast solar wind has ion speeds on the order of 750 km s^{-1} . For protons, this corresponds to an energy of $E \sim 3 \text{ keV}$. Even in

¹³An electroscope is a device used to detect electric charges. Simple electroscopes are often used as lecture demonstrations in introductory physics courses.

¹⁴Hess was a recipient of the Nobel Prize in 1936 for his discovery of cosmic rays. Hess shared the prize with Carl David Anderson, who discovered positrons; he identified them amid the myriad of cosmic-ray particles.

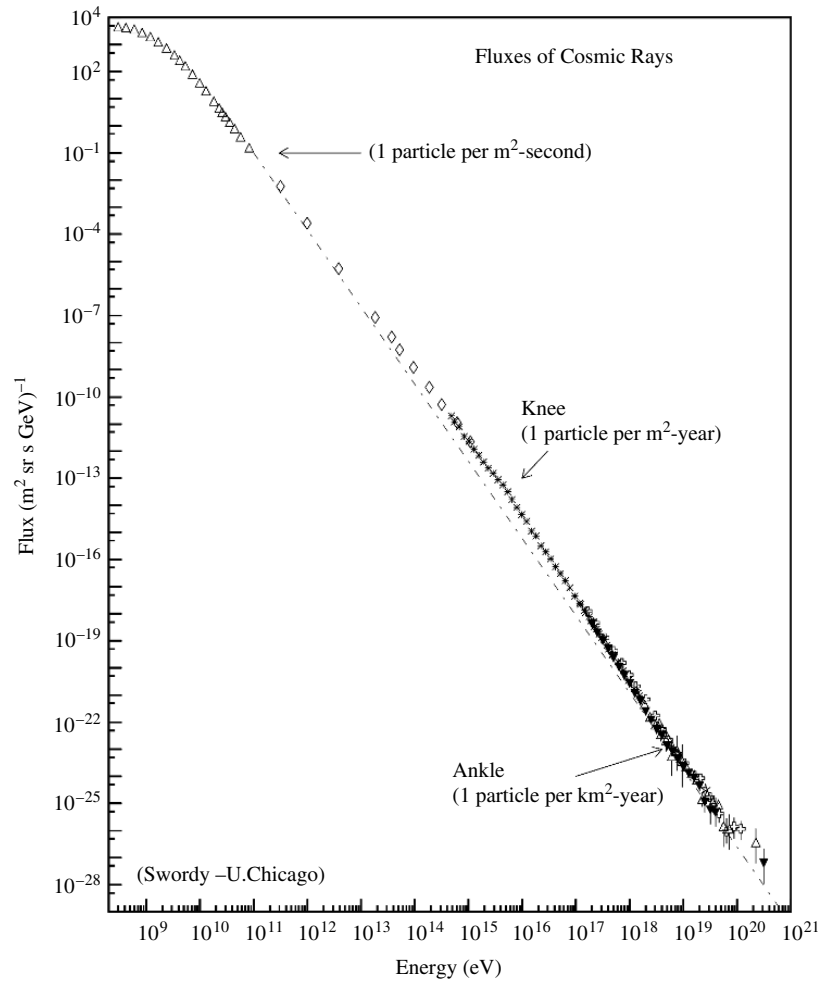


FIGURE 15.22 The flux of cosmic rays as a function of energy. (Ref: J. Cronin, T. K. Gaisser, and S. P. Swordy, *Sci. Amer.*, 276, 44, 1997.)

energetic coronal mass ejections, when protons may travel at $v \sim 0.1c$, the energies are only on the order of 10 MeV, corresponding to the lowest-energy particles shown in the nearly flat portion of the curve in Fig. 15.22.

The sources of higher-energy cosmic rays have been identified with supernovae ($E \lesssim 10^{16}$ eV; notice the “knee” in Fig. 15.22 at between 10^{15} and 10^{16} eV). At such high energies the speed of the particle is essentially c and the rest energy is negligible (for a proton, $mc^2 \simeq 937 \text{ MeV} \sim 1 \text{ GeV}$).

Consider the radius of the “orbit” of cosmic-ray particles about the magnetic field lines in the region of space in which they find themselves. From the Lorentz force equation (Eq. 11.2), if we neglect any contribution from electric fields, the force on a charged particle

in a magnetic field is given by

$$F_B = qvB$$

for the special case when the velocity of the particle is perpendicular to the magnetic field. Since the Lorentz force is always perpendicular to the direction of motion, the force is centripetal and results in a circular path of the particle around the magnetic field. This implies that

$$\frac{\gamma mv^2}{r} = qvB,$$

where γ is the Lorentz factor. Solving for r , the **Larmor radius** (or the **gyroradius**) of the orbit is given by

$$\boxed{r = \frac{\gamma mv}{qB}}. \quad (15.15)$$

Taking $v \sim c$, we find

$$r = \frac{\gamma mc^2}{qcB} = \frac{E}{qcB}. \quad (15.16)$$

Example 15.5.1. If the Larmor radius of the “orbit” significantly exceeds the size scale for the magnetic field, the particle cannot be considered to be bound to the associated system. In interstellar space, magnetic field strengths of 10^{-10} T are typical. For a proton with an energy of 10^{15} eV, the Larmor radius is

$$r = 3 \times 10^{16} \text{ m} = 1 \text{ pc}.$$

This radius is characteristic of the size of a supernova remnant, suggesting that for energies much larger than 10^{15} eV, cosmic-ray particles are not likely to be bound to a supernova remnant.

Example 15.5.1 indicates that cosmic-ray particles with energies below about 10^{15} eV are possibly associated with supernova remnants, but once their energies exceed that limit, they escape from the remnant. It has long been suggested that the shock waves associated with supernovae could be sites of acceleration of cosmic-ray particles. Enrico Fermi was the first to propose a mechanism by which supernovae could accelerate charged particles to ultra-relativistic energies. He suggested that charged particles trapped in magnetic fields can be accelerated to very high energies through successive collisions with the advancing shock wave. After absorbing energy from the shock, a particle is accelerated forward in the direction of the shock motion (think of an elastic collision of a particle with an advancing wall). However, being tied to the magnetic field in the shock’s vicinity, the particle is forced to return, only to collide with the shock again, receiving additional energy. The process is repeated many times until the particle possesses sufficient energy to escape the bonds of the

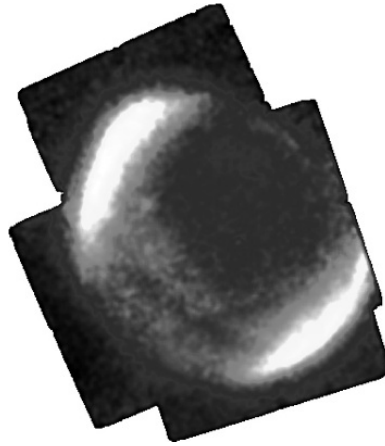


FIGURE 15.23 An X-ray image of SN 1006 obtained by ASCA. (Photo Credit: Dr. Eric V. Gotthelf, Columbia University.)

supernova's magnetic field. The power-law nature of the cosmic-ray spectrum ($F \propto E^{-\alpha}$) in Fig. 15.22 is characteristic of this nonthermal source of energy for these particles.

An X-ray image of SN 1006 (Fig. 15.23) obtained by Japan's Advanced Satellite for Cosmology and Astrophysics (ASCA) suggests strongly that supernova remnant shock fronts are indeed the source of acceleration of cosmic-ray particles with energies below about 10^{15} eV.

It remains unclear where the highest-energy cosmic rays come from. From the existence of the "knee" above 10^{15} eV in the cosmic-ray spectrum, it appears that the source of higher-energy cosmic rays is fundamentally different from the supernova source below 10^{15} eV. In addition, the "ankle" near 10^{19} eV suggests yet another source for the most extreme cosmic rays. It has been proposed that the cosmic rays with energies between the "knee" and the "ankle" may be due to acceleration in the vicinity of neutron stars or black holes. On the other hand, the most energetic cosmic rays may originate from outside our Galaxy, possible from collisions involving intergalactic shocks, or perhaps from the active regions in the centers of most galaxies where supermassive black holes are believed to reside.

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PROBLEMS

- 15.1** Estimate the Eddington limit for η Car and compare your answer with the luminosity of that star. Is your answer consistent with its behavior? Why or why not?
- 15.2** During the Great Eruption of η Car, the apparent visual magnitude reached a characteristic value of $m_V \sim 0$. Assume that the interstellar extinction to η Car is 1.7 magnitudes and that the bolometric correction is essentially zero.
- (a) Estimate the luminosity of η Car during the Great Eruption.
 - (b) Determine the total amount of photon energy liberated during the twenty years of the Great Eruption.
 - (c) If $3 M_\odot$ of material was ejected at a speed of 650 km s^{-1} , how much energy went into the kinetic energy of the ejecta?
- 15.3** The angular extent of one of the lobes of η Car is approximately $8.5''$. Assuming a constant expansion of the lobes of 650 km s^{-1} , estimate how long it has been since the Great Eruption that produced the lobes. Is this likely to be an overestimate or an underestimate? Justify your answer.
- 15.4** (a) Show that the amount of radioactive material remaining in an initially pure sample is given by Eq. (15.10).
 (b) Prove that

$$\lambda = \frac{\ln 2}{\tau_{1/2}}.$$

- 15.5** Assume that the $1 M_\odot$ core of a $10 M_\odot$ star collapses to produce a Type II supernova. Assume further that 100% of the energy released by the collapsing core is converted to neutrinos and that 1% of the neutrinos are absorbed by the overlying envelope to power the ejection of the supernova remnant. Estimate the final radius of the stellar remnant if sufficient energy is to be liberated to just barely eject the remaining $9 M_\odot$ to infinity. Be sure to state clearly any additional assumptions you make in determining your estimate of the final radius of the remnant.
- 15.6** (a) The angular size of the Crab SNR is $4' \times 2'$ and its distance from Earth is approximately 2000 pc (see Fig. 15.4). Estimate the linear dimensions of the nebula.
 (b) Using the measured expansion rate of the Crab and ignoring any accelerations since the time of the supernova explosion, estimate the age of the nebula.
- 15.7** Taking the distance to the Crab to be 2000 pc, and assuming that the absolute bolometric magnitude at maximum brightness was characteristic of a Type II supernova, estimate its peak apparent magnitude. Compare this to the maximum brightness of the planet Venus ($m \simeq -4$), which is sometimes visible in the daytime.
- 15.8** Using Eq. (12.26), make a crude estimate of the amount of time required for the homologous collapse of the inner portion of the iron core of a massive star, marking the beginning of a core-collapse supernova.
- 15.9** (a) Assuming that the light curve of a supernova is dominated by the energy released in the radioactive decay of an isotope that has a decay constant of λ , show that the slope of the light curve is given by Eq. (15.11).
 (b) Prove that Eq. (15.12) follows from Eq. (15.11).

- 15.10** If the linear decline of a supernova light curve is powered by the radioactive decay of the ejecta, find the rate of decline (in mag d^{-1}) produced by the decay of $^{56}_{27}\text{Co} \rightarrow ^{56}_{26}\text{Fe}$, with a half-life of 77.7 days.
- 15.11** The energy released during the decay of one $^{56}_{27}\text{Co}$ atom is 3.72 MeV. If $0.075 M_{\odot}$ of cobalt was produced by the decay of $^{56}_{28}\text{Ni}$ in SN 1987A, estimate the amount of energy released per second through the radioactive decay of cobalt:
- (a) just after the formation of the cobalt.
 - (b) one year after the explosion.
 - (c) Compare your answers with the light curve of SN 1987A given in Fig. 15.12.
- 15.12** The neutrino flux from SN 1987A was estimated to be $1.3 \times 10^{14} \text{ m}^{-2}$ at the location of Earth. If the average energy per neutrino was approximately 4.2 MeV, estimate the amount of energy released via neutrinos during the supernova explosion.
- 15.13** Using Eq. (10.22), estimate the gravitational binding energy of a neutron star with a mass $1.4 M_{\odot}$ and a radius of 10 km. Compare your answer with the amount of energy released in neutrinos during the collapse of the iron core of Sk –69 202 (the progenitor of SN 1987A).
- 15.14** It is estimated that there are approximately 100,000 neutron stars in the Milky Way Galaxy. Show that if the observed gamma-ray bursts are associated with neutron stars in our Galaxy, then each source *must* repeat. If you make the extreme assumption that each neutron star produces bursts, what would be the average time between bursts?
- 15.15** Consider an electron and positron that annihilate each other at the surface of a neutron star ($M = 1.4 M_{\odot}$, $R = 10 \text{ km}$), producing two gamma-ray photons of the same energy. Show that each gamma ray has an energy of at least 511 keV.
- 15.16** Suppose there are two populations of gamma-ray burst sources with energies E_1 and E_2 . Show that if the sources are distributed homogeneously throughout the universe with number densities n_1 and n_2 , respectively, then the total number of bursts observed to have a fluence $\geq S$ is proportional to $S^{-3/2}$.
- 15.17** The highest-energy cosmic-ray particle that had been recorded at the time this text was written was measured by the Fly's Eye HiRes experiment in the Utah desert in 1991. The energy of the particle was $3 \times 10^{20} \text{ eV}$.
- (a) Convert the energy of the particle to joules.
 - (b) If the particle was a baseball of mass 0.143 kg, calculate the speed of the ball.
 - (c) Convert your answer to miles per hour and compare your answer to the speed of a fast ball of the fastest major league pitchers (approximately 100 mph, or 45 m s^{-1}).
- 15.18** Using Eq. (15.16), show that cosmic-ray particles with energies of greater than 10^{19} eV are not likely to be bound to the Milky Way Galaxy. (A characteristic size scale for the Galaxy is about 30 kpc.) What about particles with energies in the range of 10^{16} eV to 10^{19} eV ?
- 15.19** Nonthermal spectra are often represented by power laws of the form

$$F = C E^{-\alpha}.$$

Figure 15.22 shows a power-law spectrum for cosmic rays. Determine the value of α in the region 10^{11} eV to the “knee,” and from the “knee” to the “ankle.”

- 15.20** Calculate the Lorentz factor for a proton with an energy of 10^{20} eV .

CHAPTER

16 The Degenerate Remnants of Stars

- 16.1 *The Discovery of Sirius B*
- 16.2 *White Dwarfs*
- 16.3 *The Physics of Degenerate Matter*
- 16.4 *The Chandrasekhar Limit*
- 16.5 *The Cooling of White Dwarfs*
- 16.6 *Neutron Stars*
- 16.7 *Pulsars*

16.1 ■ THE DISCOVERY OF SIRIUS B

In 1838 Friedrich Wilhelm Bessel (1784–1846) used the technique of stellar parallax to find the distance to the star 61 Cygni. Following this first successful measurement of a stellar distance, Bessel applied his talents to another likely candidate: Sirius, the brightest appearing star in the sky. Its parallax angle of $p'' = 0.379''$ corresponds to a distance of only 2.64 pc, or 8.61 ly (see Appendix E). Sirius's brilliance in the night sky is due in part to its proximity to Earth. As he followed the star's path through the heavens, Bessel found that it deviated slightly from a straight line. After ten years of precise observations, Bessel concluded in 1844 that Sirius is actually a binary star system. Although unable to detect the companion of the brighter star, he deduced that its orbital period was about 50 years (the modern value is 49.9 years) and predicted its position. The search was on for the unseen "Pup," the faint companion of the luminous "Dog Star."

The telescopes of Bessel's time were incapable of finding the Pup so close to the glare of its bright counterpart, and following Bessel's death in 1846 the enthusiasm for the quest waned. Finally, in 1862, Alvan Graham Clark (1832–1897), son of the prominent American lensmaker Alvan Clark (1804–1887), tested his father's new 18-inch refractor (3 inches larger than any previous instrument) on Sirius, and he promptly discovered the Pup at its predicted position. The dominant Sirius A was found to be nearly one thousand times brighter than the Pup, now called Sirius B; see Fig. 16.1. The details of their orbits about their center of mass (see Fig. 16.2 and Problem 7.4) revealed that Sirius A and Sirius B have masses of about $2.3 M_{\odot}$ and $1.0 M_{\odot}$, respectively. A more recent determination for the mass of Sirius B is $1.053 \pm 0.028 M_{\odot}$, and it is this value that we will use.

Clark's discovery of Sirius B was made near the opportune time of apastron, when the two stars were most widely separated (by just $10''$). The great difference in their luminosities ($L_A = 23.5 L_{\odot}$ and $L_B = 0.03 L_{\odot}$) makes observations at other times much more difficult.

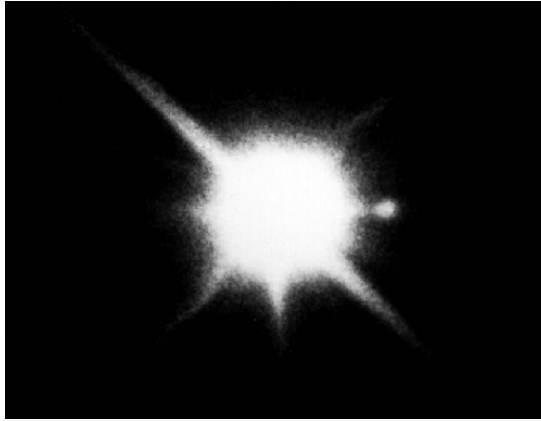


FIGURE 16.1 The white dwarf, Sirius B, beside the overexposed image of Sirius A. (Courtesy of Lick Observatory.)

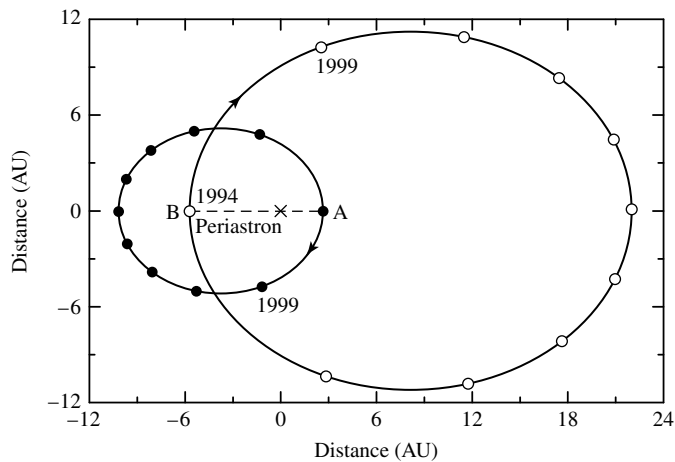


FIGURE 16.2 The orbits of Sirius A and Sirius B. The center of mass of the system is marked with an “x.”

When the next apastron arrived 50 years later, spectroscopists had developed the tools to measure the stars’ surface temperatures. From the Pup’s faint appearance, astronomers expected it to be cool and red. They were startled when Walter Adams (1876–1956), working at Mt. Wilson Observatory in 1915, discovered that, to the contrary, Sirius B is a hot, blue-white star that emits much of its energy in the ultraviolet. A modern value of the temperature of Sirius B is 27,000 K, much hotter than Sirius A’s 9910 K.

The implications for the star’s physical characteristics were astounding. Using the Stefan–Boltzmann law, Eq. (3.17), to calculate the size of Sirius B results in a radius of only $5.5 \times 10^6 \text{ m} \approx 0.008 R_{\odot}$. Sirius B has the mass of the Sun confined within a volume smaller than Earth! The average density of Sirius B is $3.0 \times 10^9 \text{ kg m}^{-3}$, and the acceler-

ation due to gravity at its surface is about $4.6 \times 10^6 \text{ m s}^{-2}$. On Earth, the pull of gravity on a teaspoon of white-dwarf material would be $1.45 \times 10^5 \text{ N}$ (over 16 tons), and on the surface of the white dwarf it would weigh 470,000 times more. This fierce gravity reveals itself in the spectrum of Sirius B; it produces an immense pressure near the surface that results in very broad hydrogen absorption lines; see Fig. 8.15.¹ Apart from these lines, its spectrum is a featureless continuum.

Astronomers first reacted to the discovery of Sirius B by dismissing the results, calling them “absurd.” However, the calculations are so simple and straightforward that this attitude soon changed to the one expressed by Eddington in 1922: “Strange objects, which persist in showing a type of spectrum entirely out of keeping with their luminosity, may ultimately teach us more than a host which radiate according to rule.” Like all sciences, astronomy advances most rapidly when confronted with exceptions to its theories.

16.2 ■ WHITE DWARFS

Obviously Sirius B is not a normal star. It is a **white dwarf**, a class of stars that have approximately the mass of the Sun and the size of Earth. Although as many as one-quarter of the stars in the vicinity of the Sun may be white dwarfs, the average characteristics of these faint stars have been difficult to determine because a complete sample has been obtained only within 10 pc of the Sun.

Classes of White Dwarf Stars

Figures 8.14 and 8.16 show that the white dwarfs occupy a narrow sliver of the H–R diagram that is roughly parallel to and below the main sequence. Although white dwarfs are typically whiter than normal stars, the name itself is something of a misnomer since they come in all colors, with surface temperatures ranging from less than 5000 K to more than 80,000 K. Their spectral type, D (for “dwarf”), has several subdivisions. The largest group (about two-thirds of the total number, including Sirius B), called **DA white dwarfs**, display only pressure-broadened hydrogen absorption lines in their spectra. Hydrogen lines are absent from the **DB white dwarfs** (8%), which show only helium absorption lines, and the **DC white dwarfs** (14%) show no lines at all—only a continuum devoid of features. The remaining types include **DQ white dwarfs**, which exhibit carbon features in their spectra, and **DZ white dwarfs** with evidence of metal lines.

Central Conditions in White Dwarfs

It is instructive to estimate the conditions at the center of a white dwarf of mass M_{wd} and radius R_{wd} , using the values for Sirius B given in the preceding section. Equation (14.5) with $r = 0$ shows that the central pressure is roughly²

$$P_c \approx \frac{2}{3} \pi G \rho^2 R_{\text{wd}}^2 \approx 3.8 \times 10^{22} \text{ N m}^{-2}, \quad (16.1)$$

¹Recall the discussion of pressure broadening in Section 9.5.

²Remember that Eq. (14.5) was obtained for the unrealistic assumption of constant density.

about 1.5 million times larger than the pressure at the center of the Sun. A crude estimate of the central temperature may be obtained from Eq. (10.68) for the radiative temperature gradient,³

$$\frac{dT}{dr} = -\frac{3}{4ac} \frac{\bar{\kappa}\rho}{T^3} \frac{L_r}{4\pi r^2}$$

or

$$\frac{T_{\text{wd}} - T_c}{R_{\text{wd}} - 0} = -\frac{3}{4ac} \frac{\bar{\kappa}\rho}{T_c^3} \frac{L_{\text{wd}}}{4\pi R_{\text{wd}}^2}.$$

Assuming that the surface temperature, T_{wd} , is much smaller than the central temperature and using $\bar{\kappa} = 0.02 \text{ m}^2 \text{ kg}^{-1}$ for electron scattering (Eq. 9.27 with $X = 0$) give

$$T_c \approx \left[\frac{3\bar{\kappa}\rho}{4ac} \frac{L_{\text{wd}}}{4\pi R_{\text{wd}}} \right]^{1/4} \approx 7.6 \times 10^7 \text{ K}.$$

Thus the central temperature of a white dwarf is several times 10^7 K .

These estimated values for a white dwarf lead directly to a surprising conclusion. Although hydrogen makes up roughly 70% of the visible mass of the universe, it cannot be present in appreciable amounts below the surface layers of a white dwarf. Otherwise, the dependence of the nuclear energy generation rates on density and temperature (see Eq. 10.46 for the pp chain and Eq. 10.58 for the CNO cycle) would produce white dwarf luminosities several orders of magnitude larger than those actually observed. Similar reasoning applied to other reaction sequences implies that thermonuclear reactions are not involved in producing the energy radiated by white dwarfs and that their centers must therefore consist of particles that are incapable of fusion at these densities and temperatures.

As was discussed in Section 13.2, white dwarfs are manufactured in the cores of low- and intermediate-mass stars (those with an initial mass below 8 or 9 $M_\odot$ on the main sequence) near the end of their lives on the asymptotic giant branch of the H–R diagram. Because any star with a helium core mass exceeding about 0.5 $M_\odot$ will undergo fusion, most white dwarfs consist primarily of completely ionized carbon and oxygen nuclei.⁴ As the aging giant expels its surface layers as a planetary nebula, the core is exposed as a white dwarf progenitor. The distribution of DA white dwarf masses is sharply peaked at 0.56 $M_\odot$, with some 80% lying between 0.42 $M_\odot$ and 0.70 $M_\odot$; see Fig. 16.3. The much larger main-sequence masses quoted earlier imply that significant amounts of mass loss occurred while on the asymptotic giant branch, involving thermal pulses and a superwind.

Spectra and Surface Composition

The exceptionally strong pull of the white dwarf's gravity is responsible for the characteristic hydrogen spectrum of DA white dwarfs. Heavier nuclei are pulled below the surface while

³As we will discuss later in Section 16.5, the assumption of a radiative temperature gradient is incorrect because the energy is actually carried outward by electron conduction. However, Eq. (10.68) is sufficient for the purpose of this estimation.

⁴Low-mass helium white dwarfs may also exist, and rare oxygen–neon–magnesium white dwarfs have been detected in a few novae.

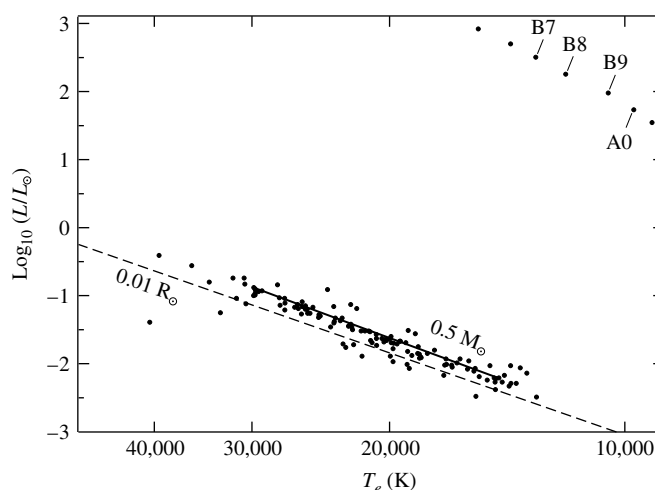


FIGURE 16.3 DA white dwarfs on an H–R diagram. A line marks the location of the $0.50 M_{\odot}$ white dwarfs, and a portion of the main sequence is at the upper right. (Data from Bergeron, Saffer, and Liebert, *Ap. J.*, 394, 228, 1992.)

the lighter hydrogen rises to the top, resulting in a thin outer layer of hydrogen covering a layer of helium on top of the carbon–oxygen core.⁵ This vertical stratification of nuclei according to their mass takes only 100 years or so in the hot atmosphere of the star. The origin of the non-DA (e.g., DB and DC) white dwarfs is not yet clear. Efficient mass-loss may occur on the asymptotic giant branch associated with the thermal pulse or superwind phases, stripping the white dwarf of nearly all of its hydrogen. Alternatively, a single white dwarf may be transformed between the DA and non-DA spectral types by convective mixing in its surface layers.⁶ For example, the helium convection zone’s penetration into a thin hydrogen layer above could change a DA into a DB white dwarf by diluting the hydrogen with additional helium.

Pulsating White Dwarfs

White dwarfs with surface temperatures of $T_e \approx 12,000$ K lie within the instability strip of the H–R diagram and pulsate with periods between 100 and 1000 s; see Fig. 8.16 and Table 14.1. These **ZZ Ceti** variables, named after the prototype discovered in 1968 by Arlo Landolt, are variable DA white dwarfs; hence they are also known as **DAV stars**. The pulsation periods correspond to nonradial g-modes that resonate within the white dwarf’s surface layers of hydrogen and helium.⁷ Because these g-modes involve almost perfectly

⁵Estimates of the relative masses of the hydrogen and helium layers range from $m(\text{H})/m(\text{He}) \approx 10^{-2}$ to 10^{-11} for DA white dwarfs.

⁶As we will see in Section 16.5, steep temperature gradients produce convection zones in the white dwarf’s surface layers.

⁷The nonradial pulsation of stars was discussed in Section 14.4. Unlike the g-modes of normal stars, shown in Fig. 14.18, the g-modes of white dwarfs are confined to their surface layers.

horizontal displacements, the radii of these compact pulsators hardly change. Their brightness variations (typically a few tenths of a magnitude) are due to temperature variations on the stars' surfaces. Since most stars will end their lives as white dwarfs, these must be the most common type of variable star in the universe, although only about seventy had been detected at the time this text was written.

Successful numerical calculations of pulsating white dwarf models were carried out by American astronomer Don Winget and others. They were able to demonstrate that it is the *hydrogen* partial ionization zone that is responsible for driving the oscillations of the ZZ Ceti stars, as mentioned in Section 14.2. These computations also confirmed the elemental stratification of white dwarf envelopes. Winget and his colleagues went on to predict that hotter DB white dwarfs should also exhibit g-mode oscillations driven by the *helium* partial ionization zone. Within a year's time, this prediction was confirmed when the first **DBV** star ($T_e \approx 27,000$ K) was discovered by Winget and his collaborators.⁸ The location of the DAV and DBV stars on the H–R diagram is shown in Fig. 16.4, along with the very hot DOV and PNNV ($T_e \approx 10^5$ K) variables that are associated with the birth of white dwarfs. (“PNN” stands for planetary nebula nuclei and the DO spectral type marks the

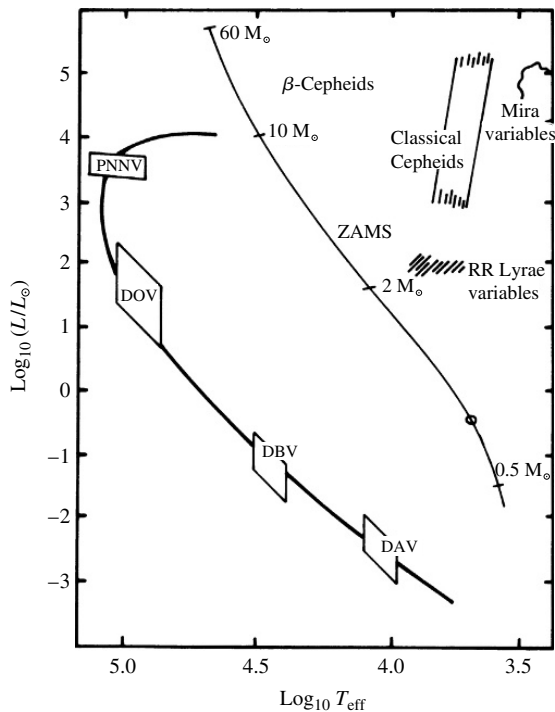


FIGURE 16.4 Compact pulsators on the H–R diagram. (Figure adapted from Winget, *Advances in Helio- and Asteroseismology*, Christensen-Dalsgaard and Frandsen (eds.), Reidel, Dordrecht, 1988.)

⁸Readers interested in this unique prediction and in the subsequent discovery of a new type of star are referred to Winget et al. (1982a,b).

transition to the white dwarf stage.) All of these stars have multiple periods, simultaneously displaying at least 3, and as many as 125, different frequencies. Astronomers are deciphering the data to obtain a detailed look at the structure of white dwarfs.

16.3 ■ THE PHYSICS OF DEGENERATE MATTER

We now delve below the surface to ask, What can support a white dwarf against the relentless pull of its gravity? It is easy to show (Problem 16.4) that normal gas and radiation pressure are completely inadequate. The answer was discovered in 1926 by the British physicist Sir Ralph Howard Fowler (1889–1944), who applied the new idea of the Pauli exclusion principle (recall Section 5.4) to the electrons within the white dwarf. The qualitative argument that follows elucidates the fundamental physics of the **electron degeneracy pressure** described by Fowler.

The Pauli Exclusion Principle and Electron Degeneracy

Any system—whether an atom of hydrogen, an oven filled with blackbody photons, or a box filled with gas particles—consists of quantum states that are identified by a set of quantum numbers. Just as the oven is filled with standing waves of electromagnetic radiation that are described by three quantum numbers (specifying the number of photons of wavelength λ traveling in the x -, y -, and z -directions), a box of gas particles is filled with standing de Broglie waves that are also described by three quantum numbers (specifying the particle's component of momentum in each of three directions). If the gas particles are fermions (such as electrons or neutrons), then the Pauli exclusion principle allows at most one fermion in each quantum state because no two fermions can have the same set of quantum numbers.

In an everyday gas at standard temperature and pressure, only one of every 10^7 quantum states is occupied by a gas particle, and the limitations imposed by the Pauli exclusion principle become insignificant. Ordinary gas has a *thermal* pressure that is related to its temperature by the ideal gas law. However, as energy is removed from the gas and its temperature falls, an increasingly large fraction of the particles are forced into the lower energy states. If the gas particles are fermions, only one particle is allowed in each state; thus all the particles cannot crowd into the ground state. Instead, as the temperature of the gas is lowered, the fermions will fill up the lowest available unoccupied states, starting with the ground state, and then successively occupy the excited states with the lowest energy. Even in the limit $T \rightarrow 0$ K, the vigorous motion of the fermions in excited states produces a pressure in the fermion gas. At zero temperature, *all* of the lower energy states and *none* of the higher energy states are occupied. Such a fermion gas is said to be completely **degenerate**.

The Fermi Energy

The maximum energy (ε_F) of any electron in a completely degenerate gas at $T = 0$ K is known as the **Fermi energy**; see Fig. 16.5. To determine this limiting energy, imagine a three-dimensional box of length L on each side. Thinking of the electrons as being standing

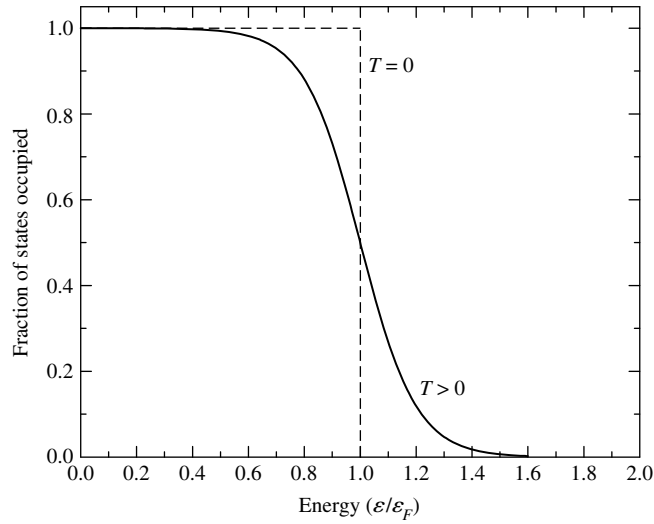


FIGURE 16.5 Fraction of states of energy ε occupied by fermions. For $T = 0$, all fermions have $\varepsilon \leq \varepsilon_F$, but for $T > 0$, some fermions have energies in excess of the Fermi energy.

waves in the box, we note that their wavelengths in each dimension are given by

$$\lambda_x = \frac{2L}{N_x}, \quad \lambda_y = \frac{2L}{N_y}, \quad \lambda_z = \frac{2L}{N_z},$$

where N_x , N_y , and N_z are integer quantum numbers associated with each dimension. Recalling that the de Broglie wavelength is related to momentum (Eq. 5.17),

$$p_x = \frac{hN_x}{2L}, \quad p_y = \frac{hN_y}{2L}, \quad p_z = \frac{hN_z}{2L}.$$

Now, the total kinetic energy of a particle can be written as

$$\varepsilon = \frac{p^2}{2m},$$

where $p^2 = p_x^2 + p_y^2 + p_z^2$. Thus,

$$\varepsilon = \frac{h^2}{8mL^2}(N_x^2 + N_y^2 + N_z^2) = \frac{h^2 N^2}{8mL^2}, \quad (16.2)$$

where $N^2 \equiv N_x^2 + N_y^2 + N_z^2$, analogous to the “distance” from the origin in “ N -space” to the point (N_x, N_y, N_z) .

The total number of electrons in the gas corresponds to the total number of unique quantum numbers, N_x , N_y , and N_z times two. The factor of two arises from the fact that electrons are spin $\frac{1}{2}$ particles, so $m_s = \pm 1/2$ implies that two electrons can have the same combination of N_x , N_y , and N_z and still possess a unique set of *four* quantum numbers (including

spin). Now, each integer coordinate in N -space (e.g., $N_x = 1, N_y = 3, N_z = 1$) corresponds to the quantum state of two electrons. With a large enough sample of electrons, they can be thought of as occupying each integer coordinate out to a radius of $N = \sqrt{N_x^2 + N_y^2 + N_z^2}$, but only for the positive octant of N -space where $N_x > 0, N_y > 0$, and $N_z > 0$. This means that the total number of electrons will be

$$N_e = 2 \left(\frac{1}{8} \right) \left(\frac{4}{3} \pi N^3 \right).$$

Solving for N yields

$$N = \left(\frac{3N_e}{\pi} \right)^{1/3}.$$

Substituting into Eq. (16.2) and simplifying, we find that the Fermi energy is given by

$$\boxed{\varepsilon_F = \frac{\hbar^2}{2m} (3\pi^2 n)^{2/3}}, \quad (16.3)$$

where m is the mass of the electron and $n \equiv N_e/L^3$ is the number of electrons per unit volume. The average energy per electron at zero temperature is $\frac{3}{5}\varepsilon_F$. (Of course the derivation above applies for any fermion, not just electrons.)

The Condition for Degeneracy

At any temperature above absolute zero, some of the states with an energy less than ε_F will become vacant as fermions use their thermal energy to occupy other, more energetic states. Although the degeneracy will not be precisely complete when $T > 0$ K, the assumption of complete degeneracy is a good approximation at the densities encountered in the interior of a white dwarf. All but the most energetic particles will have an energy less than the Fermi energy. To understand how the degree of degeneracy depends on both the temperature and the density of the white dwarf, we first express the Fermi energy in terms of the density of the electron gas. For full ionization, the number of electrons per unit volume is

$$n_e = \left(\frac{\# \text{ electrons}}{\text{nucleon}} \right) \left(\frac{\# \text{ nucleons}}{\text{volume}} \right) = \left(\frac{Z}{A} \right) \frac{\rho}{m_H}, \quad (16.4)$$

where Z and A are the number of protons and nucleons, respectively, in the white dwarf's nuclei, and m_H is the mass of a hydrogen atom.⁹ Thus the Fermi energy is proportional to the $2/3$ power of the density,

$$\varepsilon_F = \frac{\hbar^2}{2m_e} \left[3\pi^2 \left(\frac{Z}{A} \right) \frac{\rho}{m_H} \right]^{2/3}. \quad (16.5)$$

⁹The hydrogen mass is adopted as a representative mass of the proton and neutron.

Now compare the Fermi energy with the average thermal energy of an electron, $\frac{3}{2}kT$ (where k is Boltzmann's constant; see Eq. 10.17). In rough terms, if $\frac{3}{2}kT < \varepsilon_F$, then an average electron will be unable to make a transition to an unoccupied state, and the electron gas will be degenerate. That is, for a degenerate gas,

$$\frac{3}{2}kT < \frac{\hbar^2}{2m_e} \left[3\pi^2 \left(\frac{Z}{A} \right) \frac{\rho}{m_H} \right]^{2/3},$$

or

$$\frac{T}{\rho^{2/3}} < \frac{\hbar^2}{3m_e k} \left[\frac{3\pi^2}{m_H} \left(\frac{Z}{A} \right) \right]^{2/3} = 1261 \text{ K m}^2 \text{ kg}^{-2/3}$$

for $Z/A = 0.5$. Defining

$$\mathcal{D} \equiv 1261 \text{ K m}^2 \text{ kg}^{-2/3},$$

the condition for degeneracy may be written as

$$\boxed{\frac{T}{\rho^{2/3}} < \mathcal{D}.} \quad (16.6)$$

The smaller the value of $T/\rho^{2/3}$, the more degenerate the gas.

Example 16.3.1. How important is electron degeneracy at the centers of the Sun and Sirius B? At the center of the standard solar model (see Table 11.1), $T_c = 1.570 \times 10^7 \text{ K}$ and $\rho_c = 1.527 \times 10^5 \text{ kg m}^{-3}$. Then

$$\frac{T_c}{\rho_c^{2/3}} = 5500 \text{ K m}^2 \text{ kg}^{-2/3} > \mathcal{D}.$$

In the Sun, electron degeneracy is quite weak and plays a very minor role, supplying only a few tenths of a percent of the central pressure. However, as the Sun continues to evolve, electron degeneracy will become increasingly important (Fig. 16.6). As described in Section 13.2, the Sun will develop a degenerate helium core while on the red giant branch of the H–R diagram, leading eventually to a core helium flash. Later, on the asymptotic giant branch, the progenitor of a carbon–oxygen white dwarf will form in the core to be revealed when the Sun's surface layers are ejected as a planetary nebula.

For Sirius B, the values of the density and central temperature estimated above lead to

$$\frac{T_c}{\rho_c^{2/3}} = 37 \text{ K m}^2 \text{ kg}^{-2/3} \ll \mathcal{D},$$

so complete degeneracy is a valid assumption for Sirius B.

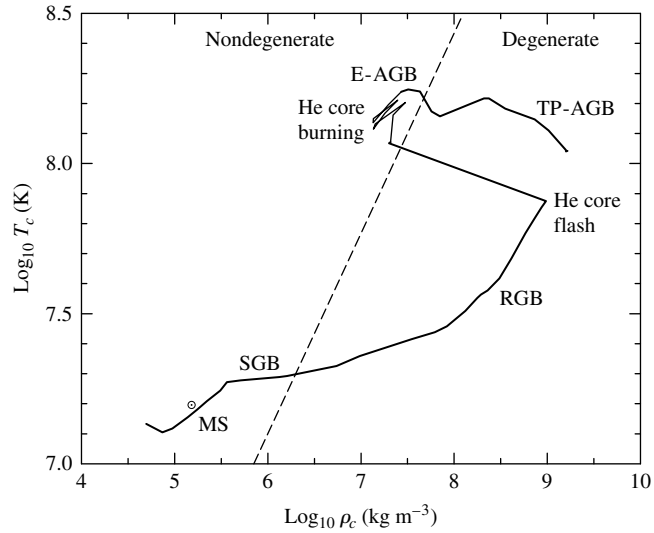


FIGURE 16.6 Degeneracy in the Sun's center as it evolves. (Data from Mazzitelli and D'Antona, *Ap. J.*, 311, 762, 1986.)

Electron Degeneracy Pressure

We now estimate the electron degeneracy pressure by combining two key ideas of quantum mechanics:

1. The Pauli exclusion principle, which allows at most one electron in each quantum state; and
2. Heisenberg's uncertainty principle in the form of Eq. (5.19),

$$\Delta x \Delta p_x \approx \hbar,$$

which requires that an electron confined to a small volume of space have a correspondingly high uncertainty in its momentum. Because the minimum value of the electron's momentum, $p_{\min}$, is approximately Δp , more closely confined electrons will have greater momenta.

When we make the unrealistic assumption that all of the electrons have the same momentum, p , Eq. (10.8) for the pressure integral becomes

$$P \approx \frac{1}{3} n_e p v, \quad (16.7)$$

where n_e is the total electron number density.

In a completely degenerate electron gas, the electrons are packed as tightly as possible, and for a uniform number density of n_e , the separation between neighboring electrons is

about $n_e^{-1/3}$. However, to satisfy the Pauli exclusion principle, the electrons must maintain their identities as different particles. That is, the uncertainty in their positions cannot be larger than their physical separation. Identifying $\Delta x \approx n_e^{-1/3}$ for the limiting case of complete degeneracy, we can use Heisenberg's uncertainty relation to estimate the momentum of an electron. In one coordinate direction,

$$p_x \approx \Delta p_x \approx \frac{\hbar}{\Delta x} \approx \hbar n_e^{1/3} \quad (16.8)$$

(see Example 5.4.2). However, in a three-dimensional gas each of the directions is equally likely, implying that

$$p_x^2 = p_y^2 = p_z^2,$$

which is just a statement of the equipartition of energy among all the coordinate directions. Therefore,

$$p^2 = p_x^2 + p_y^2 + p_z^2 = 3p_x^2,$$

or

$$p = \sqrt{3} p_x.$$

Using Eq. (16.4) for the electron number density with full ionization gives

$$p \approx \sqrt{3} \hbar \left[\left(\frac{Z}{A} \right) \frac{\rho}{m_H} \right]^{1/3}.$$

For nonrelativistic electrons, the speed is

$$\begin{aligned} v &= \frac{p}{m_e} \\ &\approx \frac{\sqrt{3} \hbar}{m_e} n_e^{1/3} \end{aligned} \quad (16.9)$$

$$\approx \frac{\sqrt{3} \hbar}{m_e} \left[\left(\frac{Z}{A} \right) \frac{\rho}{m_H} \right]^{1/3}. \quad (16.10)$$

Inserting Eqs. (16.4), (16.8), and (16.10) into Eq. (16.7) for the electron degeneracy pressure results in

$$P \approx \frac{\hbar^2}{m_e} \left[\left(\frac{Z}{A} \right) \frac{\rho}{m_H} \right]^{5/3}. \quad (16.11)$$

This is roughly a factor of two smaller than the exact expression for the pressure due to a completely degenerate, nonrelativistic electron gas P ,

$$P = \frac{(3\pi^2)^{2/3}}{5} \frac{\hbar^2}{m_e} n_e^{5/3},$$

or

$$P = \frac{(3\pi^2)^{2/3}}{5} \frac{\hbar^2}{m_e} \left[\left(\frac{Z}{A} \right) \frac{\rho}{m_H} \right]^{5/3}. \quad (16.12)$$

Using $Z/A = 0.5$ for a carbon–oxygen white dwarf, Eq. (16.12) shows that the electron degeneracy pressure available to support a white dwarf such as Sirius B is about $1.9 \times 10^{22} \text{ N m}^{-2}$, within a factor of two of the estimate of the central pressure made previously (Eq. 16.1). *Electron degeneracy pressure is responsible for maintaining hydrostatic equilibrium in a white dwarf.*

You may have noticed that Eq. (16.12) is the polytropic equation of state, $P = K\rho^{5/3}$, corresponding to $n = 1.5$. This implies that the extensive tools associated with the Lane–Emden equation (Eq. 10.110), developed beginning on page 334, can be used to study these objects. Of course, to understand them in detail requires careful numerical calculations involving the details of the complex equation of state of partially degenerate gases, nonzero temperatures, and changing compositions.

16.4 ■ THE CHANDRASEKHAR LIMIT

The requirement that degenerate electron pressure must support a white dwarf star has profound implications. In 1931, at the age of 21, the Indian physicist Subrahmanyan Chandrasekhar announced his discovery that *there is a maximum mass for white dwarfs*. In this section we will consider the physics that leads to this amazing conclusion.

The Mass–Volume Relation

The relation between the radius, R_{wd} , of a white dwarf and its mass, M_{wd} , may be found by setting the estimate of the central pressure, Eq. (16.1), equal to the electron degeneracy pressure, Eq. (16.12):

$$\frac{2}{3} \pi G \rho^2 R_{\text{wd}}^2 = \frac{(3\pi^2)^{2/3}}{5} \frac{\hbar^2}{m_e} \left[\left(\frac{Z}{A} \right) \frac{\rho}{m_H} \right]^{5/3}.$$

Using $\rho = M_{\text{wd}} / \frac{4}{3} \pi R_{\text{wd}}^3$ (assuming constant density), this leads to an estimate of the radius of the white dwarf,

$$R_{\text{wd}} \approx \frac{(18\pi)^{2/3}}{10} \frac{\hbar^2}{G m_e M_{\text{wd}}^{1/3}} \left[\left(\frac{Z}{A} \right) \frac{1}{m_H} \right]^{5/3}. \quad (16.13)$$

For a $1 M_{\odot}$ carbon–oxygen white dwarf, $R \approx 2.9 \times 10^6 \text{ m}$, too small by roughly a factor of two but an acceptable estimate. More important is the surprising implication that $M_{\text{wd}} R_{\text{wd}}^3 = \text{constant}$, or

$$M_{\text{wd}} V_{\text{wd}} = \text{constant}. \quad (16.14)$$

The volume of a white dwarf is inversely proportional to its mass, so more massive white dwarfs are actually *smaller*. This **mass–volume relation** is a result of the star deriving its support from electron degeneracy pressure. The electrons must be more closely confined to generate the larger degeneracy pressure required to support a more massive star. In fact, the mass–volume relation implies that $\rho \propto M_{\text{wd}}^2$.

According to the mass–volume relation, piling more and more mass onto a white dwarf would eventually result in shrinking the star down to zero volume as its mass becomes infinite. However, if the density exceeds about 10^9 kg m^{-3} , there is a departure from this relation. To see why this is so, use Eq. (16.10) to estimate the speed of the electrons in Sirius B:

$$v \approx \frac{\hbar}{m_e} \left[\left(\frac{Z}{A} \right) \frac{\rho}{m_H} \right]^{1/3} = 1.1 \times 10^8 \text{ m s}^{-1},$$

over one-third the speed of light! If the mass–volume relation were correct, white dwarfs a bit more massive than Sirius B would be so small and dense that their electrons would exceed the limiting value of the speed of light. This impossibility points out the dangers of ignoring the effects of relativity in our expressions for the electron speed (Eq. 16.10) and pressure (Eq. 16.11).¹⁰ Because the electrons are moving more slowly than the nonrelativistic Eq. (16.10) would indicate, there is less electron pressure available to support the star. Thus a massive white dwarf is *smaller* than predicted by the mass–volume relation. Indeed, zero volume occurs for a finite value of the mass; in other words, there is a limit to the amount of matter that can be supported by electron degeneracy pressure.

Dynamical Instability

To appreciate the effect of relativity on the stability of a white dwarf, recall that Eq. (16.12) (which is valid only for approximately $\rho < 10^9 \text{ kg m}^{-3}$) is of the polytropic form $P = K\rho^{5/3}$, where K is a constant. Comparing this with Eq. (10.86) shows that the value of the ratio of specific heats is $\gamma = 5/3$ in the nonrelativistic limit. As we discussed in Section 14.3, this means that the white dwarf is dynamically stable. If it suffers a small perturbation, it will return to its equilibrium structure instead of collapsing. However, in the extreme relativistic limit, the electron speed $v = c$ must be used instead of Eq. (16.10) to find the electron degeneracy pressure. The result is

$$P = \frac{(3\pi^2)^{1/3}}{4} \hbar c \left[\left(\frac{Z}{A} \right) \frac{\rho}{m_H} \right]^{4/3} \quad (16.15)$$

(see, for example, Problem 16.6). In this limit $\gamma = 4/3$, which corresponds to *dynamical instability*. The smallest departure from equilibrium will cause the white dwarf to collapse as electron degeneracy pressure fails.¹¹ As was explained in Section 15.3, approaching this

¹⁰It is left as an exercise to show that relativistic effects must be included for densities greater than 10^9 kg m^{-3} .

¹¹In fact, the strong gravity of the white dwarf, as described by Einstein's general theory of relativity (see Section 17.1), acts to raise the critical value of γ for dynamical instability slightly above $4/3$.

limiting case leads to the collapse of the degenerate core in an aging supergiant, resulting in a core-collapse supernova. (Note that Eq. 16.15 is a polytropic equation of state, $P = K\rho^{4/3}$, with a polytropic index of $n = 3$.)

Estimating the Chandrasekhar Limit

An approximate value for the maximum white-dwarf mass may be obtained by setting the estimate of the central pressure, Eq. (16.1) with $\rho = M_{\text{wd}}/\frac{4}{3}\pi R_{\text{wd}}^3$, equal to Eq. (16.15) with $Z/A = 0.5$. The radius of the white dwarf cancels, leaving

$$M_{\text{Ch}} \sim \frac{3\sqrt{2\pi}}{8} \left(\frac{\hbar c}{G} \right)^{3/2} \left[\left(\frac{Z}{A} \right) \frac{1}{m_H} \right]^2 = 0.44 M_{\odot} \quad (16.16)$$

for the greatest possible mass. Note that Eq. (16.16) contains three fundamental constants— $\hbar$, c , and G —representing the combined effects of quantum mechanics, relativity, and Newtonian gravitation on the structure of a white dwarf. A precise derivation with $Z/A = 0.5$ results in a value of $M_{\text{Ch}} = 1.44 M_{\odot}$, called the **Chandrasekhar limit**. Figure 16.7 shows the mass–radius relation for white dwarfs.¹² No white dwarf has been discovered with a mass exceeding the Chandrasekhar limit.¹³

It is important to emphasize that neither the nonrelativistic nor the relativistic formula for the electron degeneracy pressure developed here (Eqs. 16.12 and 16.15, respectively) contains the temperature. Unlike the gas pressure of the ideal gas law and the expression for

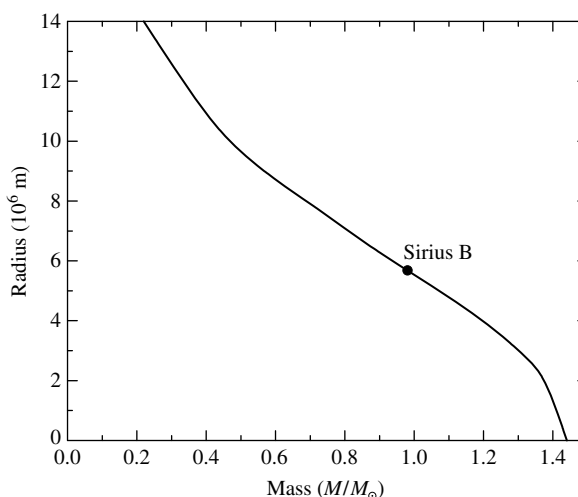


FIGURE 16.7 Radii of white dwarfs of $M_{\text{wd}} \leq M_{\text{Ch}}$ at $T = 0$ K.

¹²Figure 16.7 does not include complications such as the electrostatic attraction between the nuclei and electrons in a white dwarf, thus tending to reduce the radius slightly.

¹³It is natural to wonder about the outcome of sneaking up on the Chandrasekhar limit by adding just a bit more mass to white dwarf with very nearly $1.44 M_{\odot}$. This will be considered in Section 18.5, where Type Ia supernovae are discussed.

radiation pressure, the pressure of a completely degenerate electron gas is independent of its temperature. This has the effect of decoupling the mechanical structure of the star from its thermal properties. However, the decoupling is never perfect since $T > 0$. As a result, the correct expression for the pressure involves treating the gas as partially degenerate and relativistic, but with $v < c$. This is a challenging equation of state to deal with properly.¹⁴

We have already seen one implication of this decoupling in Section 13.2, where the helium core flash was described as the result of the independence of the mechanical and thermal behavior of the degenerate helium core of a low-mass star. When helium burning begins in the core, it proceeds without an accompanying increase in pressure that would normally expand the core and therefore restrain the rising temperature. The resulting rapid rise in temperature leads to a runaway production of nuclear energy—the helium flash—which lasts until the temperature becomes sufficiently high to remove the degeneracy of the core, allowing it to expand. On the other hand, a star may have so little mass that its core temperature never becomes high enough to initiate helium burning. The result in this case is the formation of a helium white dwarf.

16.5 ■ THE COOLING OF WHITE DWARFS

Most stars end their lives as white dwarfs. These glowing embers scattered throughout space are a galaxy's memory of its past glory. Because no fusion occurs in their interiors, white dwarfs simply cool off at an essentially constant radius as they slowly deplete their supply of thermal energy (recall Fig. 16.3). Much effort has been directed at understanding the rate at which a white dwarf cools so its lifetime and the time of its birth may be calculated. Just as paleontologists can read the history of Earth's life in the fossil record, astronomers may be able to recover the history of star formation in our Galaxy by studying the statistics of white-dwarf temperatures. This section will be devoted to a discussion of the principles involved in this stellar archaeology.

Energy Transport

First we must ask how energy is transported outward from the interior of a white dwarf. In an ordinary star, photons travel much farther than atoms do before suffering a collision that robs them of energy (recall Examples 9.2.1 and 9.2.2). As a result, photons are normally more efficient carriers of energy to the stellar surface. In a white dwarf, however, the degenerate electrons can travel long distances before losing energy in a collision with a nucleus, since the vast majority of the lower-energy electron states are already occupied. Thus, in a white dwarf, energy is carried by **electron conduction** rather than by radiation. This is so efficient that the interior of a white dwarf is nearly isothermal, with the temperature dropping significantly only in the nondegenerate surface layers. Figure 16.8 shows that a white dwarf consists of a nearly constant-temperature interior surrounded by a thin nondegenerate envelope that transfers heat less efficiently, causing the energy to leak out

¹⁴You are referred to Clayton (1983) or Hansen, Kawaler, and Trimble (2004) for a discussion of partial electron degeneracy.

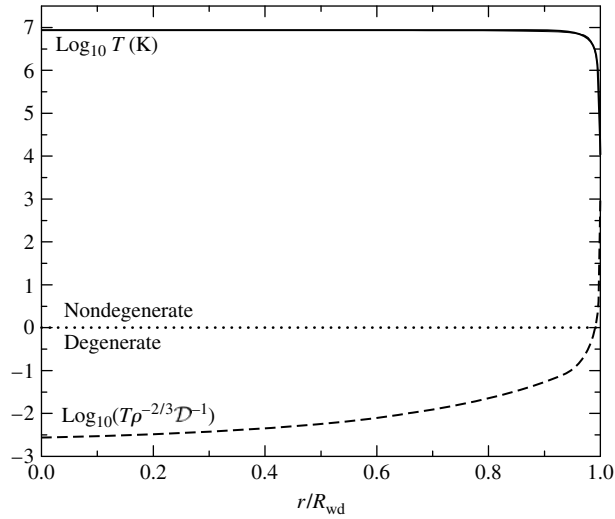


FIGURE 16.8 Temperature and degree of degeneracy in the interior of a white dwarf model. The horizontal dotted line marks the boundary between degeneracy and nondegeneracy as described by Eq. (16.6).

slowly. The steep temperature gradient near the surface creates convection zones that may alter the appearance of the white dwarf's spectrum as it cools (as described in Section 16.2).

The structure of the nondegenerate surface layers of a star is described at the beginning of Appendix L. For a white dwarf of surface luminosity L_{wd} and mass M_{wd} , Eq. (L.1) for the pressure P as a function of the temperature T in the envelope is¹⁵

$$P = \left(\frac{4}{17} \frac{16\pi ac}{3} \frac{GM_{\text{wd}}}{L_{\text{wd}}} \frac{k}{\kappa_0 \mu m_H} \right)^{1/2} T^{17/4}, \quad (16.17)$$

where κ_0 (called “ A ” in Eq. L.1) is the coefficient of the bound-free Kramers opacity law in Eq. (9.22),

$$\kappa_0 = 4.34 \times 10^{21} Z(1 + X) \text{ m}^2 \text{ kg}^{-1}.$$

Using the ideal gas law (Eq. 10.11) to replace the pressure results in a relation between the density and the temperature,

$$\rho = \left(\frac{4}{17} \frac{16\pi ac}{3} \frac{GM_{\text{wd}}}{L_{\text{wd}}} \frac{\mu m_H}{\kappa_0 k} \right)^{1/2} T^{13/4}. \quad (16.18)$$

The transition between the nondegenerate surface layers of the star and its isothermal, degenerate interior of temperature T_c is described by setting the two sides of Eq. (16.6) equal

¹⁵Equation (16.17) assumes that the envelope is in radiative equilibrium, with the energy carried outward by photons. Even when convection occurs in the surface layers of a white dwarf, it is not expected to have a large effect on the cooling.

to each other. Using this to replace the density results in an expression for the luminosity at the white dwarf's surface in terms of its interior temperature,

$$\begin{aligned} L_{\text{wd}} &= \frac{4\mathcal{D}^3}{17} \frac{16\pi ac}{3} \frac{Gm_H}{\kappa_0 k} \mu M_{\text{wd}} T_c^{7/2} \\ &= CT_c^{7/2}, \end{aligned} \quad (16.19)$$

where

$$\begin{aligned} C &\equiv \frac{4\mathcal{D}^3}{17} \frac{16\pi ac}{3} \frac{Gm_H}{\kappa_0 k} \mu M_{\text{wd}} \\ &= 6.65 \times 10^{-3} \left(\frac{M_{\text{wd}}}{M_{\odot}} \right) \frac{\mu}{Z(1+X)}. \end{aligned}$$

Note that the luminosity is proportional to $T_c^{7/2}$ (the *interior* temperature) and that it varies as the fourth power of the *effective* temperature according to the Stefan–Boltzmann law, Eq. (3.17). Thus the surface of a white dwarf cools more slowly than its isothermal interior as the star's thermal energy leaks into space.

Example 16.5.1. Equation (16.19) can be used to estimate the interior temperature of a $1 M_{\odot}$ white dwarf with $L_{\text{wd}} = 0.03 L_{\odot}$. Arbitrarily assuming values of $X = 0$, $Y = 0.9$, $Z = 0.1$ for the nondegenerate envelope (so $\mu \simeq 1.4$) results in¹⁶

$$T_c = \left[\frac{L_{\text{wd}}}{6.65 \times 10^{-3}} \left(\frac{M_{\odot}}{M_{\text{wd}}} \right) \frac{Z(1+X)}{\mu} \right]^{2/7} = 2.8 \times 10^7 \text{ K}.$$

Equating the two sides of the degeneracy condition, Eq. (16.6), shows that the density at the base of the nondegenerate envelope is about

$$\rho = \left(\frac{T_c}{\mathcal{D}} \right)^{3/2} = 3.4 \times 10^6 \text{ kg m}^{-3}.$$

This result is several orders of magnitude less than the average density of a $1 M_{\odot}$ white dwarf such as Sirius B and confirms that the envelope is indeed thin, contributing very little to the star's total mass.

The Cooling Timescale

A white dwarf's thermal energy resides primarily in the kinetic energy of its nuclei; the degenerate electrons cannot give up a significant amount of energy because nearly all of the lower energy states are already occupied. If we assume for simplicity that the composition is uniform, then the total number of nuclei in the white dwarf is equal to the star's mass, M_{wd} ,

¹⁶Because the amount of hydrogen is quite small even in a DA white dwarf, this composition is a reasonable choice for both type DA and type DB.

divided by the mass of a nucleus, Am_H . Furthermore, since the average thermal energy of a nucleus is $\frac{3}{2}kT$, the thermal energy available for radiation is

$$U = \frac{M_{\text{wd}}}{Am_H} \frac{3}{2} kT_c. \quad (16.20)$$

If we use the value of T_c from Example 16.5.1 and $A = 12$ for carbon, Eq. (16.20) gives approximately 6.0×10^{40} J. A crude estimate of the characteristic timescale for cooling, τ_{cool} , can be obtained simply by dividing the thermal energy by the luminosity. Thus

$$\tau_{\text{cool}} = \frac{U}{L_{\text{wd}}} = \frac{3}{2} \frac{M_{\text{wd}} k}{Am_H C T_c^{5/2}}, \quad (16.21)$$

which is about 5.2×10^{15} s $\approx$ 170 million years. This is an underestimate, because the cooling timescale increases as T_c decreases. The more detailed calculation that follows shows that a white dwarf spends most of its life cooling slowly with a low temperature and luminosity.

The Change in Luminosity with Time

The depletion of the internal energy provides the luminosity, so Eqs. (16.19) and (16.20) give

$$-\frac{dU}{dt} = L_{\text{wd}}$$

or

$$-\frac{d}{dt} \left(\frac{M_{\text{wd}}}{Am_H} \frac{3}{2} kT_c \right) = C T_c^{7/2}.$$

If the initial temperature of the interior is T_0 when $t = 0$, then this expression may be integrated to obtain the core temperature as a function of time:

$$T_c(t) = T_0 \left(1 + \frac{5}{3} \frac{Am_H C T_0^{5/2}}{M_{\text{wd}} k} t \right)^{-2/5} = T_0 \left(1 + \frac{5}{2} \frac{t}{\tau_0} \right)^{-2/5}, \quad (16.22)$$

where τ_0 is the timescale for cooling at the initial temperature of T_0 ; that is, $\tau_0 = \tau_{\text{cool}}$ at time t_0 . Inserting this into Eq. (16.19) shows that the luminosity of the white dwarf first declines sharply from its initial value of $L_0 = C T_0^{7/2}$ and then dims much more gradually as time passes:

$$L_{\text{wd}} = L_0 \left(1 + \frac{5}{3} \frac{Am_H C^{2/7} L_0^{5/7}}{M_{\text{wd}} k} t \right)^{-7/5} = L_0 \left(1 + \frac{5}{2} \frac{t}{\tau_0} \right)^{-7/5}. \quad (16.23)$$

The solid line in Fig. 16.9 shows the decline in the luminosity of a pure carbon $0.6 M_\odot$ white dwarf calculated from Eq. (16.23). The dashed line is a curve obtained for a sequence

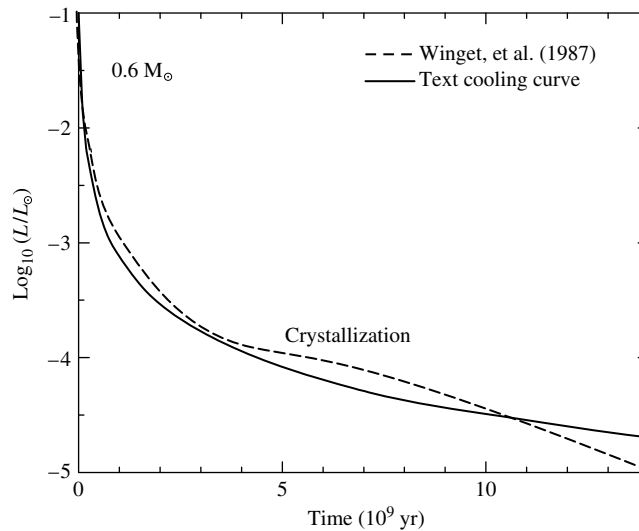


FIGURE 16.9 Theoretical cooling curves for $0.6 M_{\odot}$ white-dwarf models. [The solid line is from Eq. (16.23), and the dashed line is from Winget et al., *Ap. J. Lett.*, 315, L77, 1987.]

of more realistic white-dwarf models¹⁷ that include thin surface layers of hydrogen and helium overlying the carbon core. The insulating effect of these layers slows the cooling by about 15%. Also included are some of the intriguing phenomena that occur as the white dwarf's internal temperature drops.

Crystallization

As a white dwarf cools, it crystallizes in a gradual process that starts at the center and moves outward. The upturned “knee” in the dashed curve in Fig. 16.9 at about $L_{\text{wd}}/L_{\odot} \approx 10^{-4}$ occurs when the cooling nuclei begin settling into a crystalline lattice. The regular crystal structure is maintained by the mutual electrostatic repulsion of the nuclei; it minimizes their energy as they vibrate about their average position in the lattice. As the nuclei undergo this phase change, they release their latent heat (about kT per nucleus), slowing the star's cooling and producing the knee in the cooling curve. Later, as the white dwarf's temperature continues to drop, the crystalline lattice actually accelerates the cooling as the coherent vibration of the regularly spaced nuclei promotes further energy loss. This is reflected in the subsequent downturn in the cooling curve. Thus the ultimate monument to the lives of most stars will be a “diamond in the sky,” a cold, dark, Earth-size sphere of crystallized carbon and oxygen floating through the depths of space.¹⁸

¹⁷You are referred to Winget et al. (1987) for details of this and other cooling curves.

¹⁸Unlike a terrestrial diamond, the white dwarf's nuclei are arrayed in a body-centered cubic lattice like that of metallic sodium.

Comparing Theory with Observations

Despite the large uncertainties in the measurement of surface temperatures resulting from high surface gravities and broad spectral features,¹⁹ it is possible to observe the cooling of a pulsating white dwarf. As the star's temperature declines, its period P slowly changes according to $dP/dt \propto T^{-1}$ (approximately). Extremely precise measurements of a rapidly cooling DOV star yield a period derivative of $P/|dP/dt| = 1.4 \times 10^6$ years, in excellent agreement with the theoretical value. Measuring period changes for the more slowly cooling DBV and DAV stars are even more difficult.

This interest in an accurate calculation of the decline in a white dwarf's temperature reflects the hope of using these fossil stars as a tool for uncovering the history of star formation in our Galaxy. Figure 16.10, from Winget et al. (1987), illustrates how this might be accomplished. Each circle (both open and filled) in the figure is the observed number of white dwarfs per cubic parsec with the absolute visual magnitude given at the top of the figure. The dramatically sudden drop in the population of white dwarfs with $L_{\text{wd}}/L_{\odot} < -4.5$ is inconsistent with the assumption that stars have been forming in our Galaxy throughout the infinite past. Instead, this decline can best be explained if the first white dwarfs were formed and began cooling 9.0 ± 1.8 billion years ago. Figure 16.10 shows the theoretically expected distribution of white dwarf luminosities based on this cooling time, calculated using theoretical cooling curves similar to the one shown in Fig. 16.9

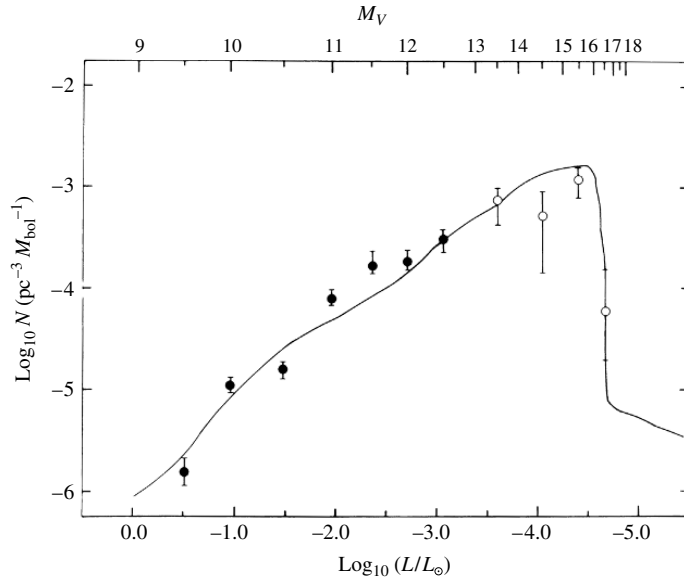


FIGURE 16.10 Observed and theoretical distribution of white-dwarf luminosities. (Figure adapted from Winget et al., *Ap. J. Lett.*, 315, L77, 1987.)

¹⁹For Sirius B, effective temperatures ranging from 27,000 K to 32,000 K are often quoted.

together with the observed distribution of white-dwarf masses. Furthermore, adding the time spent in the pre-white-dwarf stages of stellar evolution implies that star formation in the disk of our Galaxy began about 9.3 ± 2.0 billion years ago.²⁰ This time is about 3 billion years shorter than the age determined for the Milky Way's globular clusters, which formed at an earlier epoch.

16.6 ■ NEUTRON STARS

Two years after James Chadwick (1891–1974) discovered the neutron in 1932, a German astronomer and a Swiss astrophysicist, Walter Baade (1893–1960) and Fritz Zwicky (1898–1974) of Mount Wilson Observatory, proposed the existence of **neutron stars**. These two astronomers, who also coined the term *supernova*, went on to suggest that “supernovae represent the transitions from ordinary stars into neutron stars, which in their final stages consist of extremely closely packed neutrons.”

Neutron Degeneracy

Because neutron stars are formed when the degenerate core of an aging supergiant star nears the Chandrasekhar limit and collapses, we take M_{Ch} (rounded to two figures) for a typical neutron star mass. A 1.4-solar-mass neutron star would consist of $1.4 M_{\odot}/m_n \approx 10^{57}$ neutrons—in effect, a huge nucleus with a mass number of $A \approx 10^{57}$ that is held together by gravity and supported by **neutron degeneracy pressure**.²¹ It is left as an exercise to show that

$$R_{\text{ns}} \approx \frac{(18\pi)^{2/3}}{10} \frac{\hbar^2}{G M_{\text{ns}}^{1/3}} \left(\frac{1}{m_H} \right)^{8/3} \quad (16.24)$$

is the expression for the estimated neutron star radius, analogous to Eq. (16.13) for a white dwarf. For $M_{\text{ns}} = 1.4 M_{\odot}$, this yields a value of 4400 m. As we found with Eq. (16.13) for white dwarfs, this estimate is too small by a factor of about 3. That is, the actual radius of a $1.4 M_{\odot}$ neutron star lies roughly between 10 and 15 km; we will adopt a value of 10 km for the radius. As will be seen, there are many uncertainties involved in the construction of a model neutron star.

The Density of a Neutron Star

This incredibly compact stellar remnant would have an average density of $6.65 \times 10^{17} \text{ kg m}^{-3}$, greater than the typical density of an atomic nucleus, $\rho_{\text{nuc}} \approx 2.3 \times 10^{17} \text{ kg m}^{-3}$. In some sense, the neutrons in a neutron star must be “touching” one another. At the density of a neutron star, all of Earth's human inhabitants could be crowded into a cube 1.5 cm on each side.²²

²⁰Other, more recent studies have obtained similar results for the age of the *thin disk* of our Galaxy based on white-dwarf cooling times; age estimates range from 9 Gyr to 11 Gyr.

²¹Like electrons, neutrons are fermions and so are subject to the Pauli exclusion principle.

²²Astronomer Frank Shu has commented that this shows “how much of humanity is empty space”!

The pull of gravity at the surface of a neutron star is fierce. For a $1.4 M_{\odot}$ neutron star with a radius of 10 km, $g = 1.86 \times 10^{12} \text{ m s}^{-2}$, 190 billion times stronger than the acceleration of gravity at Earth's surface. An object dropped from a height of one meter would arrive at the star's surface with a speed of $1.93 \times 10^6 \text{ m s}^{-1}$ (about 4.3 million mph).

Example 16.6.1. The inadequacy of using Newtonian mechanics to describe neutron stars can be demonstrated by calculating the escape velocity at the surface. Using Eq. (2.17), we find

$$v_{\text{esc}} = \sqrt{2GM_{\text{ns}}/R_{\text{ns}}} = 1.93 \times 10^6 \text{ m s}^{-1} = 0.643c.$$

This can also be seen by considering the ratio of the Newtonian gravitational potential energy to the rest energy of an object of mass m at the star's surface:

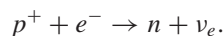
$$\frac{GM_{\text{ns}}m/R_{\text{ns}}}{mc^2} = 0.207.$$

Clearly, the effects of relativity must be included for an accurate description of a neutron star. This applies not only to Einstein's theory of special relativity, described in Chapter 4, but also to his theory of gravity, called the *general theory of relativity*, which will be considered in Section 17.1. Nevertheless, we will use both relativistic formulas and the more familiar Newtonian physics to reach qualitatively correct conclusions about neutron stars.

The Equation of State

To appreciate the exotic nature of the material constituting a neutron star and the difficulties involved in calculating the equation of state, imagine compressing the mixture of iron nuclei and degenerate electrons that make up an iron white dwarf at the center of a massive supergiant star.²³ Specifically, we are interested in the equilibrium configuration of 10^{57} nucleons (protons and neutrons), together with enough free electrons to provide zero net charge. The equilibrium arrangement is the one that involves the least energy.

Initially, at low densities the nucleons are found in iron nuclei. This is the outcome of the minimum-energy compromise between the repulsive Coulomb force between the protons and the attractive nuclear force between all of the nucleons. However, as mentioned in the discussion of the Chandrasekhar limit (Section 16.4), when $\rho \approx 10^9 \text{ kg m}^{-3}$ the electrons become relativistic. Soon thereafter, the minimum-energy arrangement of protons and neutrons changes because the energetic electrons can convert protons in the iron nuclei into neutrons by the process of electron capture (Eq. 15.6),



Because the neutron mass is slightly greater than the sum of the proton and electron masses, and the neutrino's rest-mass energy is negligible, the electron must supply the kinetic energy to make up the difference in energy; $m_n c^2 - m_p c^2 - m_e c^2 = 0.78 \text{ MeV}$.

²³Because the mechanical and thermal properties of degenerate matter are independent of one another, we will assume for convenience that $T = 0 \text{ K}$. The iron nuclei are then arranged in a crystalline lattice.

Example 16.6.2. We will obtain an estimate of the density at which the process of electron capture begins for a simple mixture of hydrogen nuclei (protons) and relativistic degenerate electrons,

$$p^+ + e^- \rightarrow n + \nu_e.$$

In the limiting case when the neutrino carries away no energy, we can equate the relativistic expression for the electron kinetic energy, Eq. (4.45), to the difference between the neutron rest energy and combined proton and electron rest energies and write

$$m_e c^2 \left(\frac{1}{\sqrt{1 - v^2/c^2}} - 1 \right) = (m_n - m_p - m_e) c^2,$$

or

$$\left(\frac{m_e}{m_n - m_p} \right)^2 = 1 - \frac{v^2}{c^2}.$$

Although Eq. (16.10) for the electron speed is strictly valid only for nonrelativistic electrons, it is accurate enough to be used in this estimate. Inserting this expression for v leads to

$$\left(\frac{m_e}{m_n - m_p} \right)^2 \approx 1 - \frac{\hbar^2}{m_e^2 c^2} \left[\left(\frac{Z}{A} \right) \frac{\rho}{m_H} \right]^{2/3}.$$

Solving for ρ shows that the density at which electron capture begins is approximately

$$\rho \approx \frac{A m_H}{Z} \left(\frac{m_e c}{\hbar} \right)^3 \left[1 - \left(\frac{m_e}{m_n - m_p} \right)^2 \right]^{3/2} \approx 2.3 \times 10^{10} \text{ kg m}^{-3},$$

using $A/Z = 1$ for hydrogen. This is in reasonable agreement with the actual value of $\rho = 1.2 \times 10^{10} \text{ kg m}^{-3}$.

We considered free protons in Example 16.6.2 to avoid the complications that arise when they are bound in heavy nuclei. A careful calculation that takes into account the surrounding nuclei and relativistic degenerate electrons, as well as the complexities of nuclear physics, reveals that the density must exceed $10^{12} \text{ kg m}^{-3}$ for the protons in ^{56}Fe nuclei to capture electrons. At still higher densities, the most stable arrangement of nucleons is one where the neutrons and protons are found in a lattice of increasingly neutron-rich nuclei so as to decrease the energy due to the Coulomb repulsion between protons. This process is known as **neutronization** and produces a sequence of nuclei such as ^{56}Fe , ^{62}Ni , ^{64}Ni , ^{66}Ni , ^{86}Kr , $\dots$, ^{118}Kr . Ordinarily, these supernumerary neutrons would revert to protons via the standard β -decay process,

$$n \rightarrow p^+ + e^- + \bar{\nu}_e.$$